

11

Systems of Nonlinear Differential Equations

EXERCISES 11.1

Autonomous Systems

1. The corresponding plane autonomous system is

$$x' = y, \quad y' = -9 \sin x.$$

If (x, y) is a critical point, $y = 0$ and $-9 \sin x = 0$. Therefore $x = \pm n\pi$ and so the critical points are $(\pm n\pi, 0)$ for $n = 0, 1, 2, \dots$.

2. The corresponding plane autonomous system is

$$x' = y, \quad y' = -2x - y^2.$$

If (x, y) is a critical point, then $y = 0$ and so $-2x - y^2 = -2x = 0$. Therefore $(0, 0)$ is the sole critical point.

3. The corresponding plane autonomous system is

$$x' = y, \quad y' = x^2 - y(1 - x^3).$$

If (x, y) is a critical point, $y = 0$ and so $x^2 - y(1 - x^3) = x^2 = 0$. Therefore $(0, 0)$ is the sole critical point.

4. The corresponding plane autonomous system is

$$x' = y, \quad y' = -4 \frac{x}{1+x^2} - 2y.$$

If (x, y) is a critical point, $y = 0$ and so $-4x/(1+x^2) - 2(0) = 0$. Therefore $x = 0$ and so $(0, 0)$ is the sole critical point.

5. The corresponding plane autonomous system is

$$x' = y, \quad y' = -x + \epsilon x^3.$$

If (x, y) is a critical point, $y = 0$ and $-x + \epsilon x^3 = 0$. Hence $x(-1 + \epsilon x^2) = 0$ and so $x = 0, \sqrt{1/\epsilon}, -\sqrt{1/\epsilon}$. The critical points are $(0, 0), (\sqrt{1/\epsilon}, 0)$ and $(-\sqrt{1/\epsilon}, 0)$.

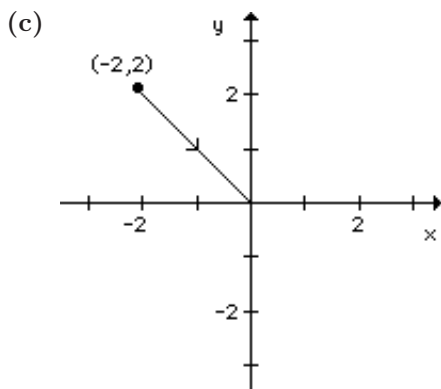
6. The corresponding plane autonomous system is

$$x' = y, \quad y' = -x + \epsilon x|x|.$$

If (x, y) is a critical point, $y = 0$ and $-x + \epsilon x|x| = x(-1 + \epsilon|x|) = 0$. Hence $x = 0, 1/\epsilon, -1/\epsilon$. The critical points are $(0, 0), (1/\epsilon, 0)$ and $(-1/\epsilon, 0)$.

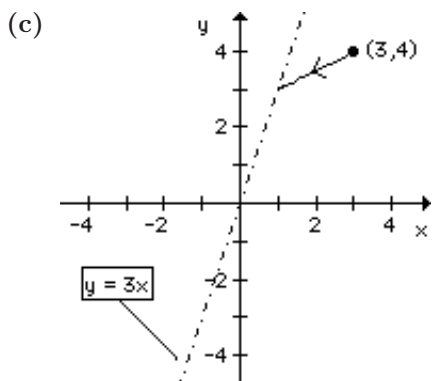
7. From $x + xy = 0$ we have $x(1+y) = 0$. Therefore $x = 0$ or $y = -1$. If $x = 0$, then, substituting into $-y - xy = 0$, we obtain $y = 0$. Likewise, if $y = -1$, $1 + x = 0$ or $x = -1$. We can conclude that $(0, 0)$ and $(-1, -1)$ are critical points of the system.

8. From $y^2 - x = 0$ we have $x = y^2$. Substituting into $x^2 - y = 0$, we obtain $y^4 - y = 0$ or $y(y^3 - 1) = 0$. It follows that $y = 0, 1$ and so $(0, 0)$ and $(1, 1)$ are the critical points of the system.
9. From $x - y = 0$ we have $y = x$. Substituting into $3x^2 - 4y = 0$ we obtain $3x^2 - 4x = x(3x - 4) = 0$. It follows that $(0, 0)$ and $(4/3, 4/3)$ are the critical points of the system.
10. From $x^3 - y = 0$ we have $y = x^3$. Substituting into $x - y^3 = 0$ we obtain $x - x^9 = 0$ or $x(1 - x^8) = 0$. Therefore $x = 0, 1, -1$ and so the critical points of the system are $(0, 0)$, $(1, 1)$, and $(-1, -1)$.
11. From $x(10 - x - \frac{1}{2}y) = 0$ we obtain $x = 0$ or $x + \frac{1}{2}y = 10$. Likewise $y(16 - y - x) = 0$ implies that $y = 0$ or $x + y = 16$. We therefore have four cases. If $x = 0$, $y = 0$ or $y = 16$. If $x + \frac{1}{2}y = 10$, we can conclude that $y(-\frac{1}{2}y + 6) = 0$ and so $y = 0, 12$. Therefore the critical points of the system are $(0, 0)$, $(0, 16)$, $(10, 0)$, and $(4, 12)$.
12. Adding the two equations we obtain $10 - 15y/(y + 5) = 0$. It follows that $y = 10$, and from $-2x + y + 10 = 0$ we can conclude that $x = 10$. Therefore $(10, 10)$ is the sole critical point of the system.
13. From $x^2 e^y = 0$ we have $x = 0$. Since $e^x - 1 = e^0 - 1 = 0$, the second equation is satisfied for an arbitrary value of y . Therefore any point of the form $(0, y)$ is a critical point.
14. From $\sin y = 0$ we have $y = \pm n\pi$. From $e^{x-y} = 1$, we can conclude that $x - y = 0$ or $x = y$. The critical points of the system are therefore $(\pm n\pi, \pm n\pi)$ for $n = 0, 1, 2, \dots$.
15. From $x(1 - x^2 - 3y^2) = 0$ we have $x = 0$ or $x^2 + 3y^2 = 1$. If $x = 0$, then substituting into $y(3 - x^2 - 3y^2) = 0$ gives $y(3 - 3y^2) = 0$. Therefore $y = 0, 1, -1$. Likewise $x^2 = 1 - 3y^2$ yields $2y = 0$ so that $y = 0$ and $x^2 = 1 - 3(0)^2 = 1$. The critical points of the system are therefore $(0, 0)$, $(0, 1)$, $(0, -1)$, $(1, 0)$, and $(-1, 0)$.
16. From $-x(4 - y^2) = 0$ we obtain $x = 0$, $y = 2$, or $y = -2$. If $x = 0$, then substituting into $4y(1 - x^2) = 0$ yields $y = 0$. Likewise $y = 2$ gives $8(1 - x^2) = 0$ or $x = 1, -1$. Finally $y = -2$ yields $-8(1 - x^2) = 0$ or $x = 1, -1$. The critical points of the system are therefore $(0, 0)$, $(1, 2)$, $(-1, 2)$, $(1, -2)$, and $(-1, -2)$.
17. (a) From Exercises 10.2, Problem 1, $x = c_1 e^{5t} - c_2 e^{-t}$ and $y = 2c_1 e^{5t} + c_2 e^{-t}$.
- (b) From $\mathbf{X}(0) = (2, -1)$ it follows that $c_1 = 0$ and $c_2 = 2$. Therefore $x = -2e^{-t}$ and $y = 2e^{-t}$.



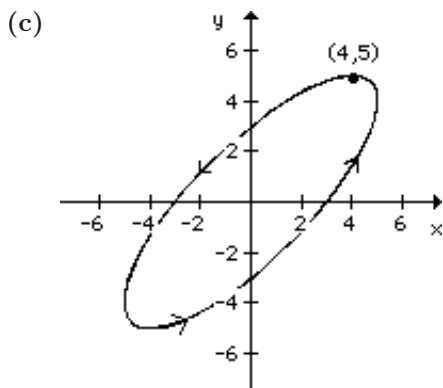
18. (a) From Exercises 10.2, Problem 6, $x = c_1 + 2c_2 e^{-5t}$ and $y = 3c_1 + c_2 e^{-5t}$, which is not periodic.
- (b) From $\mathbf{X}(0) = (3, 4)$ it follows that $c_1 = c_2 = 1$. Therefore $x = 1 + 2e^{-5t}$ and $y = 3 + e^{-5t}$ gives $y = \frac{1}{2}(x - 1) + 3$.

11.1 Autonomous Systems



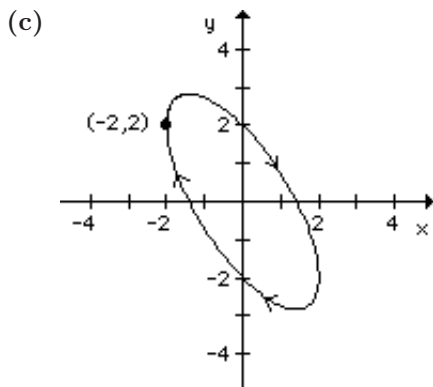
19. (a) From Exercises 10.2, Problem 37, $x = c_1(4 \cos 3t - 3 \sin 3t) + c_2(4 \sin 3t + 3 \cos 3t)$ and $y = c_1(5 \cos 3t) + c_2(5 \sin 3t)$. All solutions are one periodic with $p = 2\pi/3$.

(b) From $\mathbf{X}(0) = (4, 5)$ it follows that $c_1 = 1$ and $c_2 = 0$. Therefore $x = 4 \cos 3t - 3 \sin 3t$ and $y = 5 \cos 3t$.



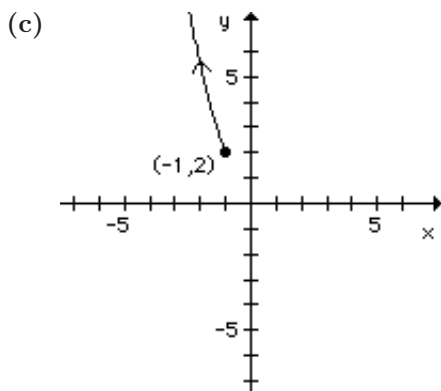
20. (a) From Exercises 10.2, Problem 34, $x = c_1(\sin t - \cos t) + c_2(-\cos t - \sin t)$ and $y = 2c_1 \cos t + 2c_2 \sin t$. All solutions are periodic with $p = 2\pi$.

(b) From $\mathbf{X}(0) = (-2, 2)$ it follows that $c_1 = c_2 = 1$. Therefore $x = -2 \cos t$ and $y = 2 \cos t + 2 \sin t$.



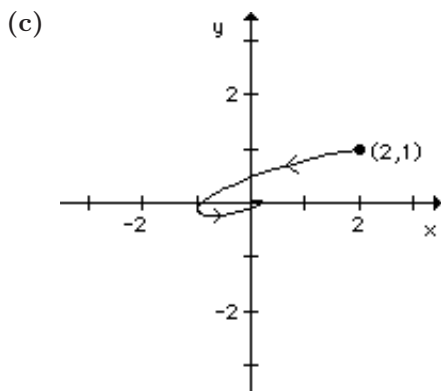
21. (a) From Exercises 10.2, Problem 35, $x = c_1(\sin t - \cos t)e^{4t} + c_2(-\sin t - \cos t)e^{4t}$ and $y = 2c_1(\cos t)e^{4t} + 2c_2(\sin t)e^{4t}$. Because of the presence of e^{4t} , there are no periodic solutions.

(b) From $\mathbf{X}(0) = (-1, 2)$ it follows that $c_1 = 1$ and $c_2 = 0$. Therefore $x = (\sin t - \cos t)e^{4t}$ and $y = 2(\cos t)e^{4t}$.



22. (a) From Exercises 10.2, Problem 38, $x = c_1 e^{-t}(2 \cos 2t - 2 \sin 2t) + c_2 e^{-t}(2 \cos 2t + 2 \sin 2t)$ and $y = c_1 e^{-t} \cos 2t + c_2 e^{-t} \sin 2t$. Because of the presence of e^{-t} , there are no periodic solutions.

(b) From $\mathbf{X}(0) = (2, 1)$ it follows that $c_1 = 1$ and $c_2 = 0$. Therefore $x = e^{-t}(2 \cos 2t - 2 \sin 2t)$ and $y = e^{-t} \cos 2t$.



23. Switching to polar coordinates,

$$\frac{dr}{dt} = \frac{1}{r} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right) = \frac{1}{r} (-xy - x^2 r^4 + xy - y^2 r^4) = -r^5$$

$$\frac{d\theta}{dt} = \frac{1}{r^2} \left(-y \frac{dx}{dt} + x \frac{dy}{dt} \right) = \frac{1}{r^2} (y^2 + xyr^4 + x^2 - xyr^4) = 1.$$

If we use separation of variables on $\frac{dr}{dt} = -r^5$ we obtain

$$r = \left(\frac{1}{4t + c_1} \right)^{1/4} \quad \text{and} \quad \theta = t + c_2.$$

Since $\mathbf{X}(0) = (4, 0)$, $r = 4$ and $\theta = 0$ when $t = 0$. It follows that $c_2 = 0$ and $c_1 = \frac{1}{256}$. The final solution can be written as

$$r = \frac{4}{\sqrt[4]{1024t + 1}}, \quad \theta = t$$

and so the solution spirals toward the origin as t increases.

24. Switching to polar coordinates,

$$\frac{dr}{dt} = \frac{1}{r} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right) = \frac{1}{r} (xy - x^2 r^2 - xy + y^2 r^2) = r^3$$

$$\frac{d\theta}{dt} = \frac{1}{r^2} \left(-y \frac{dx}{dt} + x \frac{dy}{dt} \right) = \frac{1}{r^2} (-y^2 - xyr^2 - x^2 + xyr^2) = -1.$$

11.1 Autonomous Systems

If we use separation of variables, it follows that

$$r = \frac{1}{\sqrt{-2t + c_1}} \quad \text{and} \quad \theta = -t + c_2.$$

Since $\mathbf{X}(0) = (4, 0)$, $r = 4$ and $\theta = 0$ when $t = 0$. It follows that $c_2 = 0$ and $c_1 = \frac{1}{16}$. The final solution can be written as

$$r = \frac{4}{\sqrt{1 - 32t}}, \quad \theta = -t.$$

Note that $r \rightarrow \infty$ as $t \rightarrow (\frac{1}{32})$. Because $0 \leq t \leq \frac{1}{32}$, the curve is not a spiral.

25. Switching to polar coordinates,

$$\begin{aligned} \frac{dr}{dt} &= \frac{1}{r} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right) = \frac{1}{r} [-xy + x^2(1 - r^2) + xy + y^2(1 - r^2)] = r(1 - r^2) \\ \frac{d\theta}{dt} &= \frac{1}{r^2} \left(-y \frac{dx}{dt} + x \frac{dy}{dt} \right) = \frac{1}{r^2} [y^2 - xy(1 - r^2) + x^2 + xy(1 - r^2)] = 1. \end{aligned}$$

Now $dr/dt = r - r^3$ or $(dr/dt) - r = -r^3$ is a Bernoulli differential equation. Following the procedure in Section 2.5 of the text, we let $w = r^{-2}$ so that $w' = -2r^{-3}(dr/dt)$. Therefore $w' + 2w = 2$, a linear first order differential equation. It follows that $w = 1 + c_1 e^{-2t}$ and so $r^2 = 1/(1 + c_1 e^{-2t})$. The general solution can be written as

$$r = \frac{1}{\sqrt{1 + c_1 e^{-2t}}}, \quad \theta = t + c_2.$$

If $\mathbf{X}(0) = (1, 0)$, $r = 1$ and $\theta = 0$ when $t = 0$. Therefore $c_1 = 0 = c_2$ and so $x = r \cos t = \cos t$ and $y = r \sin t = \sin t$. This solution generates the circle $r = 1$. If $\mathbf{X}(0) = (2, 0)$, $r = 2$ and $\theta = 0$ when $t = 0$. Therefore $c_1 = -3/4$, $c_2 = 0$ and so

$$r = \frac{1}{\sqrt{1 - \frac{3}{4} e^{-2t}}}, \quad \theta = t.$$

This solution spirals toward the circle $r = 1$ as t increases.

26. Switching to polar coordinates,

$$\begin{aligned} \frac{dr}{dt} &= \frac{1}{r} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right) = \frac{1}{r} \left[xy - \frac{x^2}{r}(4 - r^2) - xy - \frac{y^2}{r}(4 - r^2) \right] = r^2 - 4 \\ \frac{d\theta}{dt} &= \frac{1}{r^2} \left(-y \frac{dx}{dt} + x \frac{dy}{dt} \right) = \frac{1}{r^2} \left[-y^2 + \frac{xy}{r}(4 - r^2) - x^2 - \frac{xy}{r}(4 - r^2) \right] = -1. \end{aligned}$$

From Example 3, Section 2.2,

$$r = 2 \frac{1 + c_1 e^{4t}}{1 - c_1 e^{4t}} \quad \text{and} \quad \theta = -t + c_2.$$

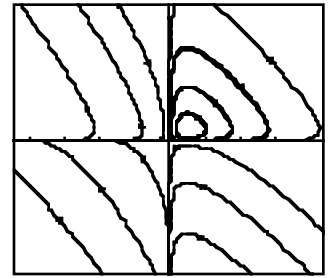
If $\mathbf{X}(0) = (1, 0)$, $r = 1$ and $\theta = 0$ when $t = 0$. It follows that $c_2 = 0$ and $c_1 = -\frac{1}{3}$. Therefore

$$r = 2 \frac{1 - \frac{1}{3} e^{4t}}{1 + \frac{1}{3} e^{4t}} \quad \text{and} \quad \theta = -t.$$

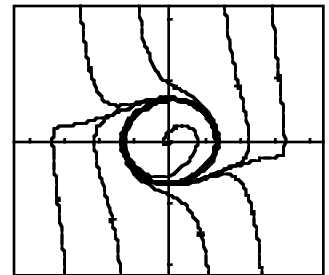
Note that $r = 0$ when $e^{4t} = 3$ or $t = (\ln 3)/4$ and $r \rightarrow -2$ as $t \rightarrow \infty$. The solution therefore approaches the circle $r = 2$. If $\mathbf{X}(0) = (2, 0)$, it follows that $c_1 = c_2 = 0$. Therefore $r = 2$ and $\theta = -t$ so that the solution generates the circle $r = 2$ traversed in the clockwise direction. Note also that the original system is not defined at $(0, 0)$ but the corresponding polar system is defined for $r = 0$. If the Runge-Kutta method is applied to the original system, the solution corresponding to $\mathbf{X}(0) = (1, 0)$ will stall at the origin.

27. The system has no critical points, so there are no periodic solutions.

28. From $x(6y - 1) = 0$ and $y(2 - 8x) = 0$ we see that $(0, 0)$ and $(1/4, 1/6)$ are critical points. From the graph we see that there are periodic solutions around $(1/4, 1/6)$.



29. The only critical point is $(0, 0)$. There appears to be a single periodic solution around $(0, 0)$.



30. The system has no critical points, so there are no periodic solutions.

31. If $\mathbf{X}(t) = (x(t), y(t))$ is a solution,

$$\frac{d}{dt} f(x(t), y(t)) = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} = QP - PQ = 0,$$

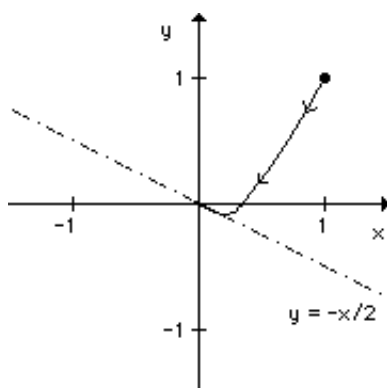
using the Chain Rule. Therefore $f(x(t), y(t)) = c$ for some constant c , and the solution lies on a level curve of the function f .

EXERCISES 11.2

Stability of Linear Systems

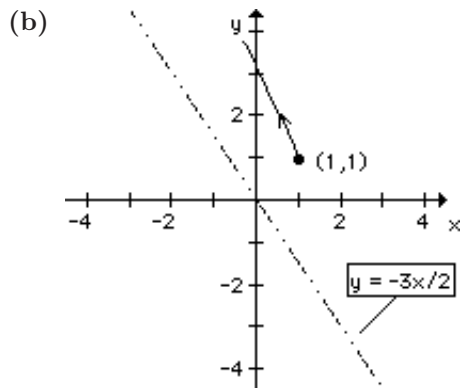
1. (a) If $\mathbf{X}(0) = \mathbf{X}_0$ lies on the line $y = 2x$, then $\mathbf{X}(t)$ approaches $(0, 0)$ along this line. For all other initial conditions, $\mathbf{X}(t)$ approaches $(0, 0)$ from the direction determined by the line $y = -x/2$.

(b)

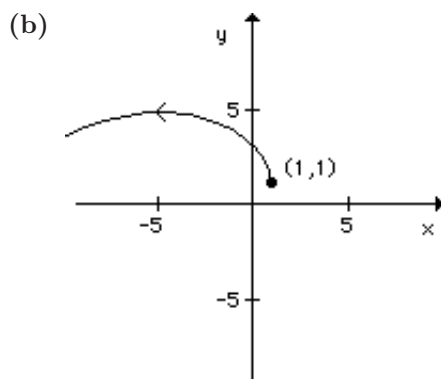


11.2 Stability of Linear Systems

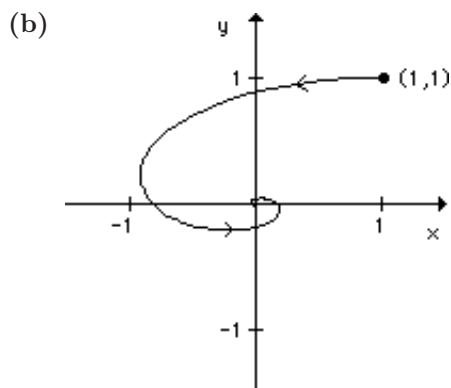
2. (a) If $\mathbf{X}(0) = \mathbf{X}_0$ lies on the line $y = -x$, then $\mathbf{X}(t)$ becomes unbounded along this line. For all other initial conditions, $\mathbf{X}(t)$ becomes unbounded and $y = -3x/2$ serves as an asymptote.



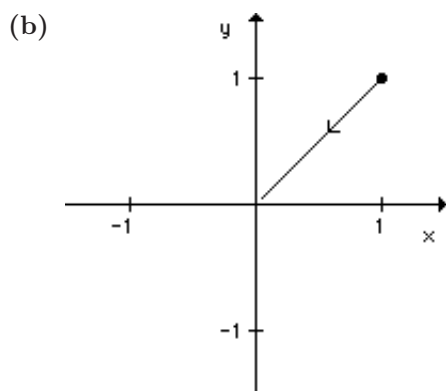
3. (a) All solutions are unstable spirals which become unbounded as t increases.



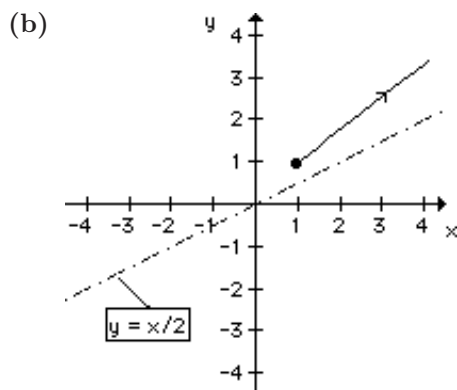
4. (a) All solutions are spirals which approach the origin.



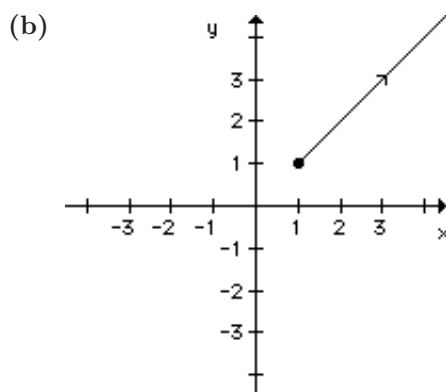
5. (a) All solutions approach $(0, 0)$ from the direction specified by the line $y = x$.



6. (a) All solutions become unbounded and $y = x/2$ serves as the asymptote.



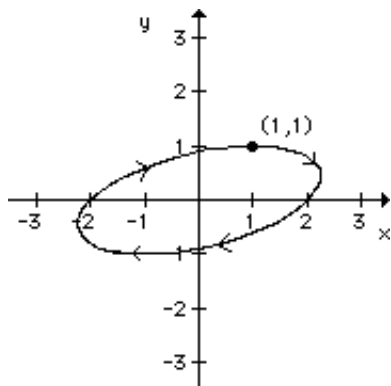
7. (a) If $\mathbf{X}(0) = \mathbf{X}_0$ lies on the line $y = 3x$, then $\mathbf{X}(t)$ approaches $(0, 0)$ along this line. For all other initial conditions, $\mathbf{X}(t)$ becomes unbounded and $y = x$ serves as the asymptote.



11.2 Stability of Linear Systems

8. (a) The solutions are ellipses which encircle the origin.

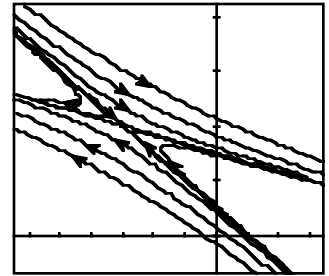
(b)



9. Since $\Delta = -41 < 0$, we can conclude from Figure 11.18 that $(0,0)$ is a saddle point.
10. Since $\Delta = 29$ and $\tau = -12$, $\tau^2 - 4\Delta > 0$ and so from Figure 11.18, $(0,0)$ is a stable node.
11. Since $\Delta = -19 < 0$, we can conclude from Figure 11.18 that $(0,0)$ is a saddle point.
12. Since $\Delta = 1$ and $\tau = -1$, $\tau^2 - 4\Delta = -3$ and so from Figure 11.18, $(0,0)$ is a stable spiral point.
13. Since $\Delta = 1$ and $\tau = -2$, $\tau^2 - 4\Delta = 0$ and so from Figure 11.18, $(0,0)$ is a degenerate stable node.
14. Since $\Delta = 1$ and $\tau = 2$, $\tau^2 - 4\Delta = 0$ and so from Figure 11.18, $(0,0)$ is a degenerate unstable node.
15. Since $\Delta = 0.01$ and $\tau = -0.03$, $\tau^2 - 4\Delta < 0$ and so from Figure 11.18, $(0,0)$ is a stable spiral point.
16. Since $\Delta = 0.0016$ and $\tau = 0.08$, $\tau^2 - 4\Delta = 0$ and so from Figure 11.18, $(0,0)$ is a degenerate unstable node.
17. $\Delta = 1 - \mu^2$, $\tau = 0$, and so we need $\Delta = 1 - \mu^2 > 0$ for $(0,0)$ to be a center. Therefore $|\mu| < 1$.
18. Note that $\Delta = 1$ and $\tau = \mu$. Therefore we need both $\tau = \mu < 0$ and $\tau^2 - 4\Delta = \mu^2 - 4 < 0$ for $(0,0)$ to be a stable spiral point. These two conditions can be written as $-2 < \mu < 0$.
19. Note that $\Delta = \mu + 1$ and $\tau = \mu + 1$ and so $\tau^2 - 4\Delta = (\mu + 1)^2 - 4(\mu + 1) = (\mu + 1)(\mu - 3)$. It follows that $\tau^2 - 4\Delta < 0$ if and only if $-1 < \mu < 3$. We can conclude that $(0,0)$ will be a saddle point when $\mu < -1$. Likewise $(0,0)$ will be an unstable spiral point when $\tau = \mu + 1 > 0$ and $\tau^2 - 4\Delta < 0$. This condition reduces to $-1 < \mu < 3$.
20. $\tau = 2\alpha$, $\Delta = \alpha^2 + \beta^2 > 0$, and $\tau^2 - 4\Delta = -4\beta^2 < 0$. If $\alpha < 0$, $(0,0)$ is a stable spiral point. If $\alpha > 0$, $(0,0)$ is an unstable spiral point. Therefore $(0,0)$ cannot be a node or saddle point.
21. $\mathbf{A}\mathbf{X}_1 + \mathbf{F} = \mathbf{0}$ implies that $\mathbf{A}\mathbf{X}_1 = -\mathbf{F}$ or $\mathbf{X}_1 = -\mathbf{A}^{-1}\mathbf{F}$. Since $\mathbf{X}_p(t) = -\mathbf{A}^{-1}\mathbf{F}$ is a particular solution, it follows from Theorem 8.6 that $\mathbf{X}(t) = \mathbf{X}_c(t) + \mathbf{X}_1$ is the general solution to $\mathbf{X}' = \mathbf{A}\mathbf{X} + \mathbf{F}$. If $\tau < 0$ and $\Delta > 0$ then $\mathbf{X}_c(t)$ approaches $(0,0)$ by Theorem 11.1(a). It follows that $\mathbf{X}(t)$ approaches $\mathbf{X}_1$ as $t \rightarrow \infty$.
22. If $bc < 1$, $\Delta = ad\hat{x}\hat{y}(1 - bc) > 0$ and $\tau^2 - 4\Delta = (a\hat{x} - d\hat{y})^2 + 4abcd\hat{x}\hat{y} > 0$. Therefore $(0,0)$ is a stable node.

23. (a) The critical point is $\mathbf{X}_1 = (-3, 4)$.

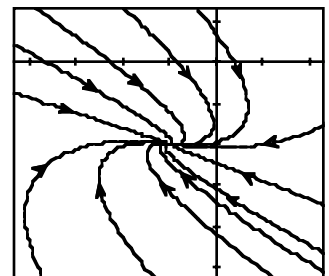
(b) From the graph, $\mathbf{X}_1$ appears to be an unstable node or a saddle point.



(c) Since $\Delta = -1$, $(0, 0)$ is a saddle point.

24. (a) The critical point is $\mathbf{X}_1 = (-1, -2)$.

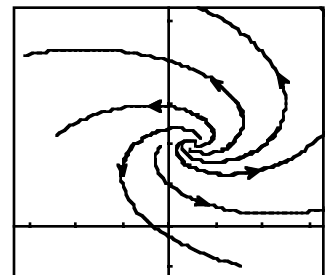
(b) From the graph, $\mathbf{X}_1$ appears to be a stable node or a degenerate stable node.



(c) Since $\tau = -16$, $\Delta = 64$, and $\tau^2 - 4\Delta = 0$, $(0, 0)$ is a degenerate stable node.

25. (a) The critical point is $\mathbf{X}_1 = (0.5, 2)$.

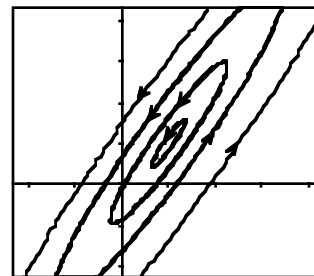
(b) From the graph, $\mathbf{X}_1$ appears to be an unstable spiral point.



(c) Since $\tau = 0.2$, $\Delta = 0.03$, and $\tau^2 - 4\Delta = -0.08$, $(0, 0)$ is an unstable spiral point.

26. (a) The critical point is $\mathbf{X}_1 = (1, 1)$.

(b) From the graph, $\mathbf{X}_1$ appears to be a center.



(c) Since $\tau = 0$ and $\Delta = 1$, $(0, 0)$ is a center.

EXERCISES 11.3

Linearization and Local Stability

1. Switching to polar coordinates,

$$\frac{dr}{dt} = \frac{1}{r} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right) = \frac{1}{r} (\alpha x^2 - \beta xy + xy^2 + \beta xy + \alpha y^2 - xy^2) = \frac{1}{r} \alpha r^2 = \alpha r.$$

Therefore $r = ce^{\alpha t}$ and so $r \rightarrow 0$ if and only if $\alpha < 0$.

2. The differential equation $dr/dt = \alpha r(5 - r)$ is a logistic differential equation. [See Section 2.8, (4) and (5).] It follows that

$$r = \frac{5}{1 + c_1 e^{-5\alpha t}} \quad \text{and} \quad \theta = -t + c_2.$$

If $\alpha > 0$, $r \rightarrow 5$ as $t \rightarrow +\infty$ and so the critical point $(0, 0)$ is unstable. If $\alpha < 0$, $r \rightarrow 0$ as $t \rightarrow +\infty$ and so $(0, 0)$ is asymptotically stable.

3. The critical points are $x = 0$ and $x = n + 1$. Since $g'(x) = k(n + 1) - 2kx$, $g'(0) = k(n + 1) > 0$ and $g'(n + 1) = -k(n + 1) < 0$. Therefore $x = 0$ is unstable while $x = n + 1$ is asymptotically stable. See Theorem 11.2.
4. Note that $x = k$ is the only critical point since $\ln(x/k)$ is not defined at $x = 0$. Since $g'(x) = -k - k \ln(x/k)$, $g'(k) = -k < 0$. Therefore $x = k$ is an asymptotically stable critical point by Theorem 11.2.
5. The only critical point is $T = T_0$. Since $g'(T) = k$, $g'(T_0) = k > 0$. Therefore $T = T_0$ is unstable by Theorem 11.2.
6. The only critical point is $v = mg/k$. Now $g(v) = g - (k/m)v$ and so $g'(v) = -k/m < 0$. Therefore $v = mg/k$ is an asymptotically stable critical point by Theorem 11.2.
7. Critical points occur at $x = \alpha, \beta$. Since $g'(x) = k(-\alpha - \beta + 2x)$, $g'(\alpha) = k(\alpha - \beta)$ and $g'(\beta) = k(\beta - \alpha)$. Since $\alpha > \beta$, $g'(\alpha) > 0$ and so $x = \alpha$ is unstable. Likewise $x = \beta$ is asymptotically stable.
8. Critical points occur at $x = \alpha, \beta, \gamma$. Since

$$g'(x) = k(\alpha - x)(-\beta - \gamma - 2x) + k(\beta - x)(\gamma - x)(-1),$$

$g'(\alpha) = -k(\beta - \alpha)(\gamma - \alpha) < 0$ since $\alpha > \beta > \gamma$. Therefore $x = \alpha$ is asymptotically stable. Similarly $g'(\beta) > 0$ and $g'(\gamma) < 0$. Therefore $x = \beta$ is unstable while $x = \gamma$ is asymptotically stable.

9. Critical points occur at $P = a/b, c$ but not at $P = 0$. Since $g'(P) = (a - bP) + (P - c)(-b)$,

$$g'(a/b) = (a/b - c)(-b) = -a + bc \quad \text{and} \quad g'(c) = a - bc.$$

Since $a < bc$, $-a + bc > 0$ and $a - bc < 0$. Therefore $P = a/b$ is unstable while $P = c$ is asymptotically stable.

10. Since $A > 0$, the only critical point is $A = K^2$. Since $g'(A) = \frac{1}{2}kKA^{-1/2} - k$, $g'(K^2) = -k/2 < 0$. Therefore $A = K^2$ is asymptotically stable.

11. The sole critical point is $(1/2, 1)$ and

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} -2y & -2x \\ 2y & 2x - 1 \end{pmatrix}.$$

Computing $\mathbf{g}'((1/2, 1))$ we find that $\tau = -2$ and $\Delta = 2$ so that $\tau^2 - 4\Delta = -4 < 0$. Therefore $(1/2, 1)$ is a stable spiral point.

12. Critical points are $(1, 0)$ and $(-1, 0)$, and

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 2x & -2y \\ 0 & 2 \end{pmatrix}.$$

At $\mathbf{X} = (1, 0)$, $\tau = 4$, $\Delta = 4$, and so $\tau^2 - 4\Delta = 0$. We can conclude that $(1, 0)$ is unstable but we are unable to classify this critical point any further. At $\mathbf{X} = (-1, 0)$, $\Delta = -4 < 0$ and so $(-1, 0)$ is a saddle point.

13. $y' = 2xy - y = y(2x - 1)$. Therefore if (x, y) is a critical point, either $x = 1/2$ or $y = 0$. The case $x = 1/2$ and $y - x^2 + 2 = 0$ implies that $(x, y) = (1/2, -7/4)$. The case $y = 0$ leads to the critical points $(\sqrt{2}, 0)$ and $(-\sqrt{2}, 0)$. We next use the Jacobian matrix

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} -2x & 1 \\ 2y & 2x - 1 \end{pmatrix}$$

to classify these three critical points. For $\mathbf{X} = (\sqrt{2}, 0)$ or $(-\sqrt{2}, 0)$, $\tau = -1$ and $\Delta < 0$. Therefore both critical points are saddle points. For $\mathbf{X} = (1/2, -7/4)$, $\tau = -1$, $\Delta = 7/2$ and so $\tau^2 - 4\Delta = -13 < 0$. Therefore $(1/2, -7/4)$ is a stable spiral point.

14. $y' = -y + xy = y(-1 + x)$. Therefore if (x, y) is a critical point, either $y = 0$ or $x = 1$. The case $y = 0$ and $2x - y^2 = 0$ implies that $(x, y) = (0, 0)$. The case $x = 1$ leads to the critical points $(1, \sqrt{2})$ and $(1, -\sqrt{2})$. We next use the Jacobian matrix

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 2 & -2y \\ y & x - 1 \end{pmatrix}$$

to classify these critical points. For $\mathbf{X} = (0, 0)$, $\Delta = -2 < 0$ and so $(0, 0)$ is a saddle point. For either $(1, \sqrt{2})$ or $(1, -\sqrt{2})$, $\tau = 2$, $\Delta = 4$, and so $\tau^2 - 4\Delta = -12$. Therefore $(1, \sqrt{2})$ and $(1, -\sqrt{2})$ are unstable spiral points.

15. Since $x^2 - y^2 = 0$, $y^2 = x^2$ and so $x^2 - 3x + 2 = (x - 1)(x - 2) = 0$. It follows that the critical points are $(1, 1)$, $(1, -1)$, $(2, 2)$, and $(2, -2)$. We next use the Jacobian

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} -3 & 2y \\ 2x & -2y \end{pmatrix}$$

to classify these four critical points. For $\mathbf{X} = (1, 1)$, $\tau = -5$, $\Delta = 2$, and so $\tau^2 - 4\Delta = 17 > 0$. Therefore $(1, 1)$ is a stable node. For $\mathbf{X} = (1, -1)$, $\Delta = -2 < 0$ and so $(1, -1)$ is a saddle point. For $\mathbf{X} = (2, 2)$, $\Delta = -4 < 0$ and so we have another saddle point. Finally, if $\mathbf{X} = (2, -2)$, $\tau = 1$, $\Delta = 4$, and so $\tau^2 - 4\Delta = -15 < 0$. Therefore $(2, -2)$ is an unstable spiral point.

16. From $y^2 - x^2 = 0$, $y = x$ or $y = -x$. The case $y = x$ leads to $(4, 4)$ and $(-1, 1)$ but the case $y = -x$ leads to $x^2 - 3x + 4 = 0$ which has no real solutions. Therefore $(4, 4)$ and $(-1, 1)$ are the only critical points. We next use the Jacobian matrix

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} y & x - 3 \\ -2x & 2y \end{pmatrix}$$

to classify these two critical points. For $\mathbf{X} = (4, 4)$, $\tau = 12$, $\Delta = 40$, and so $\tau^2 - 4\Delta < 0$. Therefore $(4, 4)$ is an unstable spiral point. For $\mathbf{X} = (-1, 1)$, $\tau = -3$, $\Delta = 10$, and so $\tau^2 - 4\Delta < 0$. It follows that $(-1, 1)$ is a stable spiral point.

17. Since $x' = -2xy = 0$, either $x = 0$ or $y = 0$. If $x = 0$, $y(1 - y^2) = 0$ and so $(0, 0)$, $(0, 1)$, and $(0, -1)$ are critical points. The case $y = 0$ leads to $x = 0$. We next use the Jacobian matrix

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} -2y & -2x \\ -1 + y & 1 + x - 3y^2 \end{pmatrix}$$

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to classify these three critical points. For $\mathbf{X} = (0, 0)$, $\tau = 1$ and $\Delta = 0$ and so the test is inconclusive. For $\mathbf{X} = (0, 1)$, $\tau = -4$, $\Delta = 4$ and so $\tau^2 - 4\Delta = 0$. We can conclude that $(0, 1)$ is a stable critical point but we are unable to classify this critical point further in this borderline case. For $\mathbf{X} = (0, -1)$, $\Delta = -4 < 0$ and so $(0, -1)$ is a saddle point.

18. We found that $(0, 0)$, $(0, 1)$, $(0, -1)$, $(1, 0)$ and $(-1, 0)$ were the critical points in Problem 15, Section 11.1. The Jacobian is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 1 - 3x^2 - 3y^2 & -6xy \\ -2xy & 3 - x^2 - 9y^2 \end{pmatrix}.$$

For $\mathbf{X} = (0, 0)$, $\tau = 4$, $\Delta = 3$ and so $\tau^2 - 4\Delta = 4 > 0$. Therefore $(0, 0)$ is an unstable node. Both $(0, 1)$ and $(0, -1)$ give $\tau = -8$, $\Delta = 12$, and $\tau^2 - 4\Delta = 16 > 0$. These two critical points are therefore stable nodes. For $\mathbf{X} = (1, 0)$ or $(-1, 0)$, $\Delta = -4 < 0$ and so saddle points occur.

19. We found the critical points $(0, 0)$, $(10, 0)$, $(0, 16)$ and $(4, 12)$ in Problem 11, Section 11.1. Since the Jacobian is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 10 - 2x - \frac{1}{2}y & -\frac{1}{2}x \\ -y & 16 - 2y - x \end{pmatrix}$$

we can classify the critical points as follows:

$\mathbf{X}$	τ	Δ	$\tau^2 - 4\Delta$	Conclusion
$(0, 0)$	26	160	36	unstable node
$(10, 0)$	-4	-60	-	saddle point
$(0, 16)$	-14	-32	-	saddle point
$(4, 12)$	-16	24	160	stable node

20. We found the sole critical point $(10, 10)$ in Problem 12, Section 11.1. The Jacobian is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} -2 & 1 \\ 2 & -1 - \frac{15}{(y+5)^2} \end{pmatrix},$$

$\mathbf{g}'((10, 10))$ has trace $\tau = -46/15$, $\Delta = 2/15$, and $\tau^2 - 4\Delta > 0$. Therefore $(0, 0)$ is a stable node.

21. The corresponding plane autonomous system is

$$\theta' = y, \quad y' = (\cos \theta - \frac{1}{2}) \sin \theta.$$

Since $|\theta| < \pi$, it follows that critical points are $(0, 0)$, $(\pi/3, 0)$ and $(-\pi/3, 0)$. The Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ \cos 2\theta - \frac{1}{2} \cos \theta & 0 \end{pmatrix}$$

and so at $(0, 0)$, $\tau = 0$ and $\Delta = -1/2$. Therefore $(0, 0)$ is a saddle point. For $\mathbf{X} = (\pm\pi/3, 0)$, $\tau = 0$ and $\Delta = 3/4$. It is not possible to classify either critical point in this borderline case.

22. The corresponding plane autonomous system is

$$x' = y, \quad y' = -x + \left(\frac{1}{2} - 3y^2\right)y - x^2.$$

If (x, y) is a critical point, $y = 0$ and so $-x - x^2 = -x(1 + x) = 0$. Therefore $(0, 0)$ and $(-1, 0)$ are the only two critical points. We next use the Jacobian matrix

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ -1 - 2x & \frac{1}{2} - 9y^2 \end{pmatrix}$$

to classify these critical points. For $\mathbf{X} = (0, 0)$, $\tau = 1/2$, $\Delta = 1$, and $\tau^2 - 4\Delta < 0$. Therefore $(0, 0)$ is an unstable spiral point. For $\mathbf{X} = (-1, 0)$, $\tau = 1/2$, $\Delta = -1$ and so $(-1, 0)$ is a saddle point.

23. The corresponding plane autonomous system is

$$x' = y, \quad y' = x^2 - y(1 - x^3)$$

and the only critical point is $(0, 0)$. Since the Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ 2x + 3x^2y & x^3 - 1 \end{pmatrix},$$

$\tau = -1$ and $\Delta = 0$, and we are unable to classify the critical point in this borderline case.

24. The corresponding plane autonomous system is

$$x' = y, \quad y' = -\frac{4x}{1+x^2} - 2y$$

and the only critical point is $(0, 0)$. Since the Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ -4\frac{1-x^2}{(1+x^2)^2} & -2 \end{pmatrix},$$

$\tau = -2$, $\Delta = 4$, $\tau^2 - 4\Delta = -12$, and so $(0, 0)$ is a stable spiral point.

25. In Problem 5, Section 11.1, we showed that $(0, 0)$, $(\sqrt{1/\epsilon}, 0)$ and $(-\sqrt{1/\epsilon}, 0)$ are the critical points. We will use the Jacobian matrix

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ -1 + 3\epsilon x^2 & 0 \end{pmatrix}$$

to classify these three critical points. For $\mathbf{X} = (0, 0)$, $\tau = 0$ and $\Delta = 1$ and we are unable to classify this critical point. For $(\pm\sqrt{1/\epsilon}, 0)$, $\tau = 0$ and $\Delta = -2$ and so both of these critical points are saddle points.

26. In Problem 6, Section 11.1, we showed that $(0, 0)$, $(1/\epsilon, 0)$, and $(-1/\epsilon, 0)$ are the critical points. Since $D_x x|x| = 2|x|$, the Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ 2\epsilon|x| - 1 & 0 \end{pmatrix}.$$

For $\mathbf{X} = (0, 0)$, $\tau = 0$, $\Delta = 1$ and we are unable to classify this critical point. For $(\pm 1/\epsilon, 0)$, $\tau = 0$, $\Delta = -1$, and so both of these critical points are saddle points.

27. The corresponding plane autonomous system is

$$x' = y, \quad y' = -\frac{(\beta + \alpha^2 y^2)x}{1 + \alpha^2 x^2}$$

and the Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ \frac{(\beta + \alpha y^2)(\alpha^2 x^2 - 1)}{(1 + \alpha^2 x^2)^2} & \frac{-2\alpha^2 yx}{1 + \alpha^2 x^2} \end{pmatrix}.$$

For $\mathbf{X} = (0, 0)$, $\tau = 0$ and $\Delta = \beta$. Since $\beta < 0$, we can conclude that $(0, 0)$ is a saddle point.

28. From $x' = -\alpha x + xy = x(-\alpha + y) = 0$, either $x = 0$ or $y = \alpha$. If $x = 0$, then $1 - \beta y = 0$ and so $y = 1/\beta$. The case $y = \alpha$ implies that $1 - \beta\alpha - x^2 = 0$ or $x^2 = 1 - \alpha\beta$. Since $\alpha\beta > 1$, this equation has no real solutions. It follows that $(0, 1/\beta)$ is the unique critical point. Since the Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} -\alpha + y & x \\ -2x & -\beta \end{pmatrix},$$

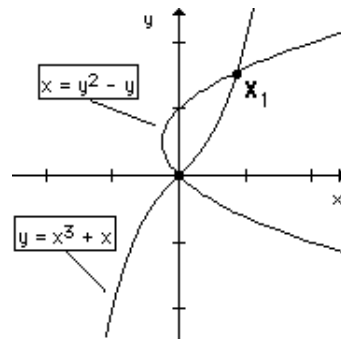
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$\tau = -\alpha - \beta + \frac{1}{\beta} = -\beta + \frac{1-\alpha\beta}{\beta} < 0$ and $\Delta = \alpha\beta - 1 > 0$. Therefore $(0, 1/\beta)$ is a stable critical point.

29. (a) The graphs of $-x + y - x^3 = 0$ and $-x - y + y^2 = 0$ are shown in the figure. The Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} -1 - 3x^2 & 1 \\ -1 & -1 + 2y \end{pmatrix}.$$

For $\mathbf{X} = (0, 0)$, $\tau = -2$, $\Delta = 2$, $\tau^2 - 4\Delta = -4$, and so $(0, 0)$ is a stable spiral point.



- (b) For $\mathbf{X}_1$, $\Delta = -6.07 < 0$ and so a saddle point occurs at $\mathbf{X}_1$.

30. (a) The corresponding plane autonomous system is

$$x' = y, \quad y' = \epsilon \left(y - \frac{1}{3}y^3 \right) - x$$

and so the only critical point is $(0, 0)$. Since the Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ -1 & \epsilon(1 - y^2) \end{pmatrix},$$

$\tau = \epsilon$, $\Delta = 1$, and so $\tau^2 - 4\Delta = \epsilon^2 - 4$ at the critical point $(0, 0)$.

- (b) When $\tau = \epsilon > 0$, $(0, 0)$ is an unstable critical point.
- (c) When $\epsilon < 0$ and $\tau^2 - 4\Delta = \epsilon^2 - 4 < 0$, $(0, 0)$ is a stable spiral point. These two requirements can be written as $-2 < \epsilon < 0$.
- (d) When $\epsilon = 0$, $x'' + x = 0$ and so $x = c_1 \cos t + c_2 \sin t$. Therefore all solutions are periodic (with period 2π) and so $(0, 0)$ is a center.
31. The differential equation $dy/dx = y'/x' = -2x^3/y$ can be solved by separating variables. It follows that $y^2 + x^4 = c$. If $\mathbf{X}(0) = (x_0, 0)$ where $x_0 > 0$, then $c = x_0^4$ so that $y^2 = x_0^4 - x^4$. Therefore if $-x_0 < x < x_0$, $y^2 > 0$ and so there are two values of y corresponding to each value of x . Therefore the solution $\mathbf{X}(t)$ with $\mathbf{X}(0) = (x_0, 0)$ is periodic and so $(0, 0)$ is a center.
32. The differential equation $dy/dx = y'/x' = (x^2 - 2x)/y$ can be solved by separating variables. It follows that $y^2/2 = (x^3/3) - x^2 + c$ and since $\mathbf{X}(0) = (x(0), x'(0)) = (1, 0)$, $c = \frac{2}{3}$. Therefore

$$\frac{y^2}{2} = \frac{x^3 - 3x^2 + 2}{3} = \frac{(x-1)(x^2 - 2x - 2)}{3}.$$

But $(x-1)(x^2 - 2x - 2) > 0$ for $1 - \sqrt{3} < x < 1$ and so each x in this interval has 2 corresponding values of y . therefore $\mathbf{X}(t)$ is a periodic solution.

33. (a) $x' = 2xy = 0$ implies that either $x = 0$ or $y = 0$. If $x = 0$, then from $1 - x^2 + y^2 = 0$, $y^2 = -1$ and there are no real solutions. If $y = 0$, $1 - x^2 = 0$ and so $(1, 0)$ and $(-1, 0)$ are critical points. The Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 2y & 2x \\ -2x & 2y \end{pmatrix}$$

and so $\tau = 0$ and $\Delta = 4$ at either $\mathbf{X} = (1, 0)$ or $(-1, 0)$. We obtain no information about these critical points in this borderline case.

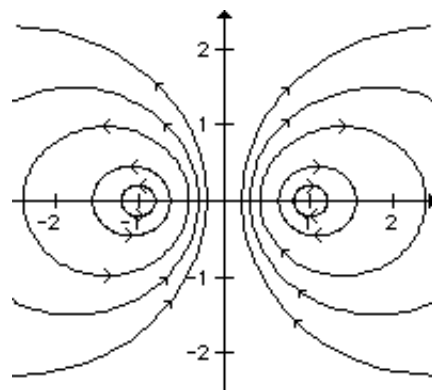
(b) The differential equation is

$$\frac{dy}{dx} = \frac{y'}{x'} = \frac{1 - x^2 + y^2}{2xy}$$

or

$$2xy \frac{dy}{dx} = 1 - x^2 + y^2.$$

Letting $\mu = y^2/x$, it follows that $d\mu/dx = (1/x^2) - 1$ and so $\mu = -(1/x) - x + 2c$. Therefore $y^2/x = -(1/x) - x + 2c$ which can be put in the form $(x - c)^2 + y^2 = c^2 - 1$. The solution curves are shown and so both $(1, 0)$ and $(-1, 0)$ are centers.



34. (a) The differential equation is $dy/dx = y'/x' = (-x - y^2)/y = -(x/y) - y$ and so $dy/dx + y = -xy^{-1}$.
- (b) Let $w = y^{1-n} = y^2$. It follows that $dw/dx + 2w = -2x$, a linear first order differential equation whose solution is $y^2 = w = ce^{-2x} + (\frac{1}{2} - x)$. Since $x(0) = \frac{1}{2}$ and $y(0) = x'(0) = 0$, $0 = c$ and so $y^2 = \frac{1}{2} - x$, a parabola with vertex at $(1/2, 0)$. Therefore the solution $\mathbf{X}(t)$ with $\mathbf{X}(0) = (1/2, 0)$ is not periodic.
35. The differential equation is $dy/dx = y'/x' = (x^3 - x)/y$ and so $y^2/2 = x^4/4 - x^2/2 + c$ or $y^2 = x^4/2 - x^2 + c_1$. Since $x(0) = 0$ and $y(0) = x'(0) = v_0$, it follows that $c_1 = v_0^2$ and so

$$y^2 = \frac{1}{2}x^4 - x^2 + v_0^2 = \frac{(x^2 - 1)^2 + 2v_0^2 - 1}{2}.$$

The x -intercepts on this graph satisfy

$$x^2 = 1 \pm \sqrt{1 - 2v_0^2}$$

and so we must require that $1 - 2v_0^2 \geq 0$ (or $|v_0| \leq \frac{1}{2}\sqrt{2}$) for real solutions to exist. If $x_0^2 = 1 - \sqrt{1 - 2v_0^2}$ and $-x_0 < x < x_0$, then $(x^2 - 1)^2 + 2v_0^2 - 1 > 0$ and so there are two corresponding values of y . Therefore $\mathbf{X}(t)$ with $\mathbf{X}(0) = (0, v_0)$ is periodic provided that $|v_0| \leq \frac{1}{2}\sqrt{2}$.

36. The corresponding plane autonomous system is

$$x' = y, \quad y' = \epsilon x^2 - x + 1$$

and so the critical points must satisfy $y = 0$ and

$$x = \frac{1 \pm \sqrt{1 - 4\epsilon}}{2\epsilon}.$$

Therefore we must require that $\epsilon \leq \frac{1}{4}$ for real solutions to exist. We will use the Jacobian matrix

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ 2\epsilon x - 1 & 0 \end{pmatrix}$$

to attempt to classify $((1 \pm \sqrt{1 - 4\epsilon})/2\epsilon, 0)$ when $\epsilon \leq 1/4$. Note that $\tau = 0$ and $\Delta = \mp\sqrt{1 - 4\epsilon}$. For $\mathbf{X} = ((1 + \sqrt{1 - 4\epsilon})/2\epsilon, 0)$ and $\epsilon < 1/4$, $\Delta < 0$ and so a saddle point occurs. For $\mathbf{X} = ((1 - \sqrt{1 - 4\epsilon})/2\epsilon, 0)$, $\Delta \geq 0$ and we are not able to classify this critical point using linearization.

37. The corresponding plane autonomous system is

$$x' = y, \quad y' = -\frac{\alpha}{L}x - \frac{\beta}{L}x^3 - \frac{R}{L}y$$

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where $x = q$ and $y = q'$. If $\mathbf{X} = (x, y)$ is a critical point, $y = 0$ and $-\alpha x - \beta x^3 = -x(\alpha + \beta x^2) = 0$. If $\beta > 0$, $\alpha + \beta x^2 = 0$ has no real solutions and so $(0, 0)$ is the only critical point. Since

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ \frac{-\alpha - 3\beta x^2}{L} & -\frac{R}{L} \end{pmatrix},$$

$\tau = -R/L < 0$ and $\Delta = \alpha/L > 0$. Therefore $(0, 0)$ is a stable critical point. If $\beta < 0$, $(0, 0)$ and $(\pm\hat{x}, 0)$, where $\hat{x}^2 = -\alpha/\beta$ are critical points. At $\mathbf{X}(\pm\hat{x}, 0)$, $\tau = -R/L < 0$ and $\Delta = -2\alpha/L < 0$. Therefore both critical points are saddles.

38. If we let $dx/dt = y$, then $dy/dt = -x^3 - x$. From this we obtain the first-order differential equation

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = -\frac{x^3 + x}{y}.$$

Separating variables and integrating we obtain

$$\int y \, dy = -\int (x^3 + x) \, dx$$

and

$$\frac{1}{2}y^2 = -\frac{1}{4}x^4 - \frac{1}{2}x^2 + c_1.$$

Completing the square we can write the solution as $y^2 = -\frac{1}{2}(x^2 + 1)^2 + c_2$. If $\mathbf{X}(0) = (x_0, 0)$, then $c_2 = \frac{1}{2}(x_0^2 + 1)^2$ and so

$$\begin{aligned} y^2 &= -\frac{1}{2}(x^2 + 1)^2 + \frac{1}{2}(x_0^2 + 1)^2 = \frac{x_0^4 + 2x_0^2 + 1 - x^4 - 2x^2 - 1}{2} \\ &= \frac{(x_0^2 + x^2)(x_0^2 - x^2) + 2(x_0^2 - x^2)}{2} = \frac{(x_0^2 + x^2 + 2)(x_0^2 - x^2)}{2}. \end{aligned}$$

Note that $y = 0$ when $x = -x_0$. In addition, the right-hand side is positive for $-x_0 < x < x_0$, and so there are two corresponding values of y for each x between $-x_0$ and x_0 . The solution $\mathbf{X} = \mathbf{X}(t)$ that satisfies $\mathbf{X}(0) = (x_0, 0)$ is therefore periodic, and so $(0, 0)$ is a center.

39. (a) Letting $x = \theta$ and $y = x'$ we obtain the system $x' = y$ and $y' = 1/2 - \sin x$. Since $\sin \pi/6 = \sin 5\pi/6 = 1/2$ we see that $(\pi/6, 0)$ and $(5\pi/6, 0)$ are critical points of the system.
(b) The Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ -\cos x & 0 \end{pmatrix}$$

and so

$$\mathbf{A}_1 = \mathbf{g}'((\pi/6, 0)) = \begin{pmatrix} 0 & 1 \\ -\sqrt{3}/2 & 0 \end{pmatrix} \quad \text{and} \quad \mathbf{A}_2 = \mathbf{g}'((5\pi/6, 0)) = \begin{pmatrix} 0 & 1 \\ \sqrt{3}/2 & 0 \end{pmatrix}.$$

Since $\det \mathbf{A}_1 > 0$ and the trace of $\mathbf{A}_1$ is 0, no conclusion can be drawn regarding the critical point $(\pi/6, 0)$.

Since $\det \mathbf{A}_2 < 0$, we see that $(5\pi/6, 0)$ is a saddle point.

- (c) From the system in part (a) we obtain the first-order differential equation

$$\frac{dy}{dx} = \frac{1/2 - \sin x}{y}.$$

Separating variables and integrating we obtain

$$\int y \, dy = \int \left(\frac{1}{2} - \sin x \right) dx$$

and

$$\frac{1}{2}y^2 = \frac{1}{2}x + \cos x + c_1$$

or

$$y^2 = x + 2\cos x + c_2.$$

For x_0 near $\pi/6$, if $\mathbf{X}(0) = (x_0, 0)$ then $c_2 = -x_0 - 2\cos x_0$ and $y^2 = x + 2\cos x - x_0 - 2\cos x_0$. Thus, there are two values of y for each x in a sufficiently small interval around $\pi/6$. Therefore $(\pi/6, 0)$ is a center.

40. (a) Writing the system as $x' = x(x^3 - 2y^3)$ and $y' = y(2x^3 - y^3)$ we see that $(0, 0)$ is a critical point. Setting $x^3 - 2y^3 = 0$ we have $x^3 = 2y^3$ and $2x^3 - y^3 = 4y^3 - y^3 = 3y^3$. Thus, $(0, 0)$ is the only critical point of the system.

- (b) From the system we obtain the first-order differential equation

$$\frac{dy}{dx} = \frac{2x^3y - y^4}{x^4 - 2xy^3}$$

or

$$(2x^3y - y^4) dx + (2xy^3 - x^4) dy = 0$$

which is homogeneous. If we let $y = ux$ it follows that

$$(2x^4u - x^4u^4) dx + (2x^4u^3 - x^4)(u dx + x du) = 0$$

$$x^4u(1 + u^3) dx + x^5(2u^3 - 1) du = 0$$

$$\frac{1}{x} dx + \frac{2u^3 - 1}{u(u^3 + 1)} du = 0$$

$$\frac{1}{x} dx + \left(\frac{1}{u+1} - \frac{1}{u} + \frac{2u-1}{u^2-u+1} \right) du = 0.$$

Integrating gives

$$\ln|x| + \ln|u+1| - \ln|u| + \ln|u^2-u+1| = c_1$$

or

$$x \left(\frac{u+1}{u} \right) (u^2 - u + 1) = c_2$$

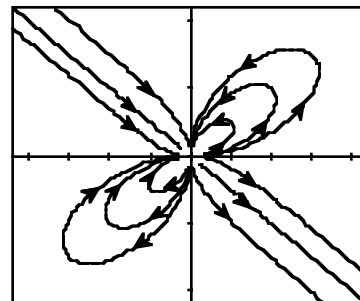
$$x \left(\frac{y+x}{y} \right) \left(\frac{y^2}{x^2} - \frac{y}{x} + 1 \right) = c_2$$

$$(xy + x^2)(y^2 - xy + x^2) = c_2x^2y$$

$$xy^3 + x^4 = c_2x^2y$$

$$x^3 + y^2 = 3c_3xy.$$

- (c) We see from the graph that $(0, 0)$ is unstable. It is not possible to classify the critical point as a node, saddle, center, or spiral point.



EXERCISES 11.4

Autonomous Systems as Mathematical Models

1. We are given that $x(0) = \theta(0) = \pi/3$ and $y(0) = \theta'(0) = w_0$. Since $y^2 = (2g/l) \cos x + c$, $w_0^2 = (2g/l) \cos(\pi/3) + c = g/l + c$ and so $c = w_0^2 - g/l$. Therefore

$$y^2 = \frac{2g}{l} \left(\cos x - \frac{1}{2} + \frac{l}{2g} w_0^2 \right)$$

and the x -intercepts occur where $\cos x = 1/2 - (l/2g)w_0^2$ and so $1/2 - (l/2g)w_0^2$ must be greater than -1 for solutions to exist. This condition is equivalent to $|w_0| < \sqrt{3g/l}$.

2. (a) Since $y^2 = (2g/l) \cos x + c$, $x(0) = \theta(0) = \theta_0$ and $y(0) = \theta'(0) = 0$, $c = -(2g/l) \cos \theta_0$ and so $y^2 = 2g(\cos \theta - \cos \theta_0)/l$. When $\theta = -\theta_0$, $y^2 = 2g[\cos(-\theta_0) - \cos \theta_0]/l = 0$. Therefore $y = d\theta/dt = 0$ when $\theta = \theta_0$.

- (b) Since $y = d\theta/dt$ and θ is decreasing between the time when $\theta = \theta_0$, $t = 0$, and $\theta = -\theta_0$, that is, $t = T$,

$$\frac{d\theta}{dt} = -\sqrt{\frac{2g}{l}} \sqrt{\cos \theta - \cos \theta_0}.$$

Therefore

$$\frac{dt}{d\theta} = -\sqrt{\frac{l}{2g}} \frac{1}{\sqrt{\cos \theta - \cos \theta_0}}$$

and so

$$T = -\sqrt{\frac{l}{2g}} \int_{\theta=\theta_0}^{\theta=-\theta_0} \frac{1}{\sqrt{\cos \theta - \cos \theta_0}} d\theta = \sqrt{\frac{l}{2g}} \int_{-\theta_0}^{\theta_0} \frac{1}{\sqrt{\cos \theta - \cos \theta_0}} d\theta.$$

3. The corresponding plane autonomous system is

$$x' = y, \quad y' = -g \frac{f'(x)}{1 + [f'(x)]^2} - \frac{\beta}{m} y$$

and

$$\frac{\partial}{\partial x} \left(-g \frac{f'(x)}{1 + [f'(x)]^2} - \frac{\beta}{m} y \right) = -g \frac{(1 + [f'(x)]^2)f''(x) - f'(x)2f'(x)f''(x)}{(1 + [f'(x)]^2)^2}.$$

If $\mathbf{X}_1 = (x_1, y_1)$ is a critical point, $y_1 = 0$ and $f'(x_1) = 0$. The Jacobian at this critical point is therefore

$$\mathbf{g}'(\mathbf{X}_1) = \begin{pmatrix} 0 & 1 \\ -gf''(x_1) & -\frac{\beta}{m} \end{pmatrix}.$$

4. When $\beta = 0$ the Jacobian matrix is

$$\begin{pmatrix} 0 & 1 \\ -gf''(x_1) & 0 \end{pmatrix}$$

which has complex eigenvalues $\lambda = \pm \sqrt{gf''(x_1)} i$. The approximating linear system with $x'(0) = 0$ has solution

$$x(t) = x(0) \cos \sqrt{gf''(x_1)} t$$

and period $2\pi/\sqrt{gf''(x_1)}$. Therefore $p \approx 2\pi/\sqrt{gf''(x_1)}$ for the actual solution.

5. (a) If $f(x) = x^2/2$, $f'(x) = x$ and so

$$\frac{dy}{dx} = \frac{y'}{x'} = -g \frac{x}{1+x^2} \frac{1}{y}.$$

We can separate variables to show that $y^2 = -g \ln(1+x^2) + c$. But $x(0) = x_0$ and $y(0) = x'(0) = v_0$. Therefore $c = v_0^2 + g \ln(1+x_0^2)$ and so

$$y^2 = v_0^2 - g \ln \left(\frac{1+x^2}{1+x_0^2} \right).$$

Now

$$v_0^2 - g \ln \left(\frac{1+x^2}{1+x_0^2} \right) \geq 0 \quad \text{if and only if} \quad x^2 \leq e^{v_0^2/g}(1+x_0^2) - 1.$$

Therefore, if $|x| \leq [e^{v_0^2/g}(1+x_0^2) - 1]^{1/2}$, there are two values of y for a given value of x and so the solution is periodic.

- (b) Since $z = x^2/2$, the maximum height occurs at the largest value of x on the cycle. From (a), $x_{\max} = [e^{v_0^2/g}(1+x_0^2) - 1]^{1/2}$ and so

$$z_{\max} = \frac{x_{\max}^2}{2} = \frac{1}{2}[e^{v_0^2/g}(1+x_0^2) - 1].$$

6. (a) If $f(x) = \cosh x$, $f'(x) = \sinh x$ and $[f'(x)]^2 + 1 = \sinh^2 x + 1 = \cosh^2 x$. Therefore

$$\frac{dy}{dx} = \frac{y'}{x'} = -g \frac{\sinh x}{\cosh^2 x} \frac{1}{y}.$$

We can separate variables to show that $y^2 = 2g/\cosh x + c$. But $x(0) = x_0$ and $y(0) = x'(0) = v_0$. Therefore $c = v_0^2 - (2g/\cosh x_0)$ and so

$$y^2 = \frac{2g}{\cosh x} - \frac{2g}{\cosh x_0} + v_0^2.$$

Now

$$\frac{2g}{\cosh x} - \frac{2g}{\cosh x_0} + v_0^2 \geq 0 \quad \text{if and only if} \quad \cosh x \leq \frac{2g \cosh x_0}{2g - v_0^2 \cosh x_0}$$

and the solution to this inequality is an interval $[-a, a]$. Therefore each x in $(-a, a)$ has two corresponding values of y and so the solution is periodic.

- (b) Since $z = \cosh x$, the maximum height occurs at the largest value of x on the cycle. From (a), $x_{\max} = a$ where $\cosh a = 2g \cosh x_0 / (2g - v_0^2 \cosh x_0)$. Therefore

$$z_{\max} = \frac{2g \cosh x_0}{2g - v_0^2 \cosh x_0}.$$

7. If $x_m < x_1 < x_n$, then $F(x_1) > F(x_m) = F(x_n)$. Letting $x = x_1$,

$$G(y) = \frac{c_0}{F(x_1)} = \frac{F(x_m)G(a/b)}{F(x_1)} < G(a/b).$$

Therefore from Property (2) in the discussion preceding Example 3 in this section of the text, $G(y) = c_0/F(x_1)$ has two solutions y_1 and y_2 that satisfy $y_1 < a/b < y_2$.

8. From Property (1) in the discussion preceding Example 3 in this section of the text, when $y = a/b$, x_n is taken on at some time t . From Property (3), if $x > x_n$ there is no corresponding value of y . Therefore the maximum number of predators is x_n and x_n occurs when $y = a/b$.

11.4 Autonomous Systems as Mathematical Models

9. (a) In the Lotka-Volterra Model the average number of predators is d/c and the average number of prey is a/b . But

$$x' = -ax + bxy - \epsilon_1 x = -(a + \epsilon_1)x + bxy$$

$$y' = -cxy + dy - \epsilon_2 y = -cxy + (d - \epsilon_2)y$$

and so the new critical point in the first quadrant is $(d/c - \epsilon_2/c, a/b + \epsilon_1/b)$.

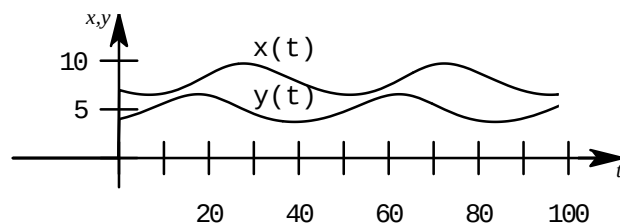
- (b) The average number of predators $d/c - \epsilon_2/c$ has decreased while the average number of prey $a/b + \epsilon_1/b$ has increased. The fishery science model is consistent with Volterra's principle.

10. (a) Solving

$$x(-0.1 + 0.02y) = 0$$

$$y(0.2 - 0.025x) = 0$$

in the first quadrant we obtain the critical point $(8, 5)$. The graphs are plotted using $x(0) = 7$ and $y(0) = 4$.



- (b) The graph in part (a) was obtained using **NDSolve** in *Mathematica*. We see that the period is around 40. Since $x(0) = 7$, we use the **FindRoot** equation solver in *Mathematica* to approximate the solution of $x(t) = 7$ for t near 40. From this we see that the period is more closely approximated by $t = 44.65$.

11. Solving

$$x(20 - 0.4x - 0.3y) = 0$$

$$y(10 - 0.1y - 0.3x) = 0$$

we see that critical points are $(0, 0)$, $(0, 100)$, $(50, 0)$, and $(20, 40)$. The Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0.08(20 - 0.8x - 0.3y) & -0.024x \\ -0.018y & 0.06(10 - 0.2y - 0.3x) \end{pmatrix}$$

and so

$$\mathbf{A}_1 = \mathbf{g}'((0, 0)) = \begin{pmatrix} 1.6 & 0 \\ 0 & 0.6 \end{pmatrix}$$

$$\mathbf{A}_2 = \mathbf{g}'((0, 100)) = \begin{pmatrix} -0.8 & 0 \\ -1.8 & -0.6 \end{pmatrix}$$

$$\mathbf{A}_3 = \mathbf{g}'((50, 0)) = \begin{pmatrix} -1.6 & -1.2 \\ 0 & -0.3 \end{pmatrix}$$

$$\mathbf{A}_4 = \mathbf{g}'((20, 40)) = \begin{pmatrix} -0.64 & -0.48 \\ -0.72 & -0.24 \end{pmatrix}.$$

Since $\det(\mathbf{A}_1) = \Delta_1 = 0.96 > 0$, $\tau = 2.2 > 0$, and $\tau_1^2 - 4\Delta_1 = 1 > 0$, we see that $(0, 0)$ is an unstable node. Since $\det(\mathbf{A}_2) = \Delta_2 = 0.48 > 0$, $\tau = -1.4 < 0$, and $\tau_2^2 - 4\Delta_2 = 0.04 > 0$, we see that $(0, 100)$ is a stable node. Since $\det(\mathbf{A}_3) = \Delta_3 = 0.48 > 0$, $\tau = -1.9 < 0$, and $\tau_3^2 - 4\Delta_3 = 1.69 > 0$, we see that $(50, 0)$ is a stable node. Since $\det(\mathbf{A}_4) = -0.192 < 0$ we see that $(20, 40)$ is a saddle point.

12. $\Delta = r_1 r_2$, $\tau = r_1 + r_2$ and $\tau^2 - 4\Delta = (r_1 + r_2)^2 - 4r_1 r_2 = (r_1 - r_2)^2$. Therefore when $r_1 \neq r_2$, $(0, 0)$ is an unstable node.
13. For $\mathbf{X} = (K_1, 0)$, $\tau = -r_1 + r_2[1 - (K_1 \alpha_{21}/K_2)]$ and $\Delta = -r_1 r_2[1 - (K_1 \alpha_{21}/K_2)]$. If we let $c = 1 - K_1 \alpha_{21}/K_2$, $\tau^2 - 4\Delta = (cr_2 + r_1)^2 > 0$. Now if $k_1 > K_2/\alpha_{21}$, $c < 0$ and so $\tau < 0$, $\Delta > 0$. Therefore $(K_1, 0)$ is a stable node. If $K_1 < K_2/\alpha_{21}$, $c > 0$ and so $\Delta < 0$. In this case $(K_1, 0)$ is a saddle point.
14. $(\hat{x}, \hat{y})$ is a stable node if and only if $K_1/\alpha_{12} > K_2$ and $K_2/\alpha_{21} > K_1$. [See Figure 11.38(a) in the text.] From Problem 12, $(0, 0)$ is an unstable node and from Problem 13, since $K_1 < K_2/\alpha_{21}$, $(K_1, 0)$ is a saddle point. Finally, when $K_2 < K_1/\alpha_{12}$, $(0, K_2)$ is a saddle point. This is Problem 12 with the roles of 1 and 2 interchanged. Therefore $(0, 0)$, $(K_1, 0)$, and $(0, K_2)$ are unstable.

15. $K_1/\alpha_{12} < K_2 < K_1\alpha_{21}$ and so $\alpha_{12}\alpha_{21} > 1$. Therefore $\Delta = (1 - \alpha_{12}\alpha_{21})\hat{x}\hat{y}r_1r_2/K_1K_2 < 0$ and so $(\hat{x}, \hat{y})$ is a saddle point.

16. (a) The corresponding plane autonomous system is

$$x' = y, \quad y' = \frac{-g}{l} \sin x - \frac{\beta}{ml} y$$

and so critical points must satisfy both $y = 0$ and $\sin x = 0$. Therefore $(\pm n\pi, 0)$ are critical points.

- (b) The Jacobian matrix

$$\begin{pmatrix} 0 & 1 \\ -\frac{g}{l} \cos x & -\frac{\beta}{ml} \end{pmatrix}$$

has trace $\tau = -\beta/ml$ and determinant $\Delta = g/l > 0$ at $(0, 0)$. Therefore

$$\tau^2 - 4\Delta = \frac{\beta^2}{m^2 l^2} - 4 \frac{g}{l} = \frac{\beta^2 - 4g l m^2}{m^2 l^2}.$$

We can conclude that $(0, 0)$ is a stable spiral point provided $\beta^2 - 4g l m^2 < 0$ or $\beta < 2m\sqrt{gl}$.

17. (a) The corresponding plane autonomous system is

$$x' = y, \quad y' = -\frac{\beta}{m} y|y| - \frac{k}{m} x$$

and so a critical point must satisfy both $y = 0$ and $x = 0$. Therefore $(0, 0)$ is the unique critical point.

- (b) The Jacobian matrix is

$$\begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{\beta}{m} 2|y| \end{pmatrix}$$

and so $\tau = 0$ and $\Delta = k/m > 0$. Therefore $(0, 0)$ is a center, stable spiral point, or an unstable spiral point. Physical considerations suggest that $(0, 0)$ must be asymptotically stable and so $(0, 0)$ must be a stable spiral point.

18. (a) The magnitude of the frictional force between the bead and the wire is $\mu(mg \cos \theta)$ for some $\mu > 0$. The component of this frictional force in the x -direction is

$$(\mu mg \cos \theta) \cos \theta = \mu mg \cos^2 \theta.$$

But

$$\cos \theta = \frac{1}{\sqrt{1 + [f'(x)]^2}} \quad \text{and so} \quad \mu mg \cos^2 \theta = \frac{\mu mg}{1 + [f'(x)]^2}.$$

It follows from Newton's Second Law that

$$mx'' = -mg \frac{f'(x)}{1 + [f'(x)]^2} - \beta x' + mg \frac{\mu}{1 + [f'(x)]^2}$$

and so

$$x'' = g \frac{\mu - f'(x)}{1 + [f'(x)]^2} - \frac{\beta}{m} x'.$$

- (b) A critical point (x, y) must satisfy $y = 0$ and $f'(x) = \mu$. Therefore critical points occur at $(x_1, 0)$ where $f'(x_1) = \mu$. The Jacobian matrix of the plane autonomous system is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ g \frac{(1 + [f'(x)]^2)(-f''(x)) - (\mu - f'(x))2f'(x)f''(x)}{(1 + [f'(x)]^2)^2} & -\frac{\beta}{m} \end{pmatrix}$$

11.4 Autonomous Systems as Mathematical Models

and so at a critical point $\mathbf{X}_1$,

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ \frac{-gf''(x_1)}{1+\mu^2} & -\frac{\beta}{m} \end{pmatrix}.$$

Therefore $\tau = -\beta/m < 0$ and $\Delta = gf''(x_1)/(1+\mu^2)$. When $f''(x_1) < 0$, $\Delta < 0$ and so a saddle point occurs. When $f''(x_1) > 0$ and

$$\tau^2 - 4\Delta = \frac{\beta^2}{m^2} - 4g \frac{f''(x_1)}{1+\mu^2} < 0,$$

$(x_1, 0)$ is a stable spiral point. This condition can also be written as

$$\beta^2 < 4gm^2 \frac{f''(x_1)}{1+\mu^2}.$$

19. We have $dy/dx = y'/x' = -f(x)/y$ and so, using separation of variables,

$$\frac{y^2}{2} = -\int_0^x f(\mu) d\mu + c \quad \text{or} \quad y^2 + 2F(x) = c.$$

We can conclude that for a given value of x there are at most two corresponding values of y . If $(0, 0)$ were a stable spiral point there would exist an x with more than two corresponding values of y . Note that the condition $f(0) = 0$ is required for $(0, 0)$ to be a critical point of the corresponding plane autonomous system $x' = y$, $y' = -f(x)$.

20. (a) $x' = x(-a + by) = 0$ implies that $x = 0$ or $y = a/b$. If $x = 0$, then, from

$$-cxy + \frac{r}{K} y(K - y) = 0,$$

$y = 0$ or K . Therefore $(0, 0)$ and $(0, K)$ are critical points. If $\hat{y} = a/b$, then

$$\hat{y} \left[-cx + \frac{r}{K}(K - \hat{y}) \right] = 0.$$

The corresponding value of x , $x = \hat{x}$, therefore satisfies the equation $c\hat{x} = r(K - \hat{y})/K$.

- (b) The Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} -a + by & bx \\ -cy & -cx + \frac{r}{K}(K - 2y) \end{pmatrix}$$

and so at $\mathbf{X}_1 = (0, 0)$, $\Delta = -ar < 0$. For $\mathbf{X}_1 = (0, K)$, $\Delta = n(Kb - a) = -rb(K - a/b)$. Since we are given that $K > a/b$, $\Delta < 0$ in this case. Therefore $(0, 0)$ and $(0, K)$ are each saddle points. For $\mathbf{X}_1 = (\hat{x}, \hat{y})$ where $\hat{y} = a/b$ and $c\hat{x} = r(K - \hat{y})/K$, we can write the Jacobian matrix as

$$\mathbf{g}'((\hat{x}, \hat{y})) = \begin{pmatrix} 0 & b\hat{x} \\ -c\hat{y} & -\frac{r}{K}\hat{y} \end{pmatrix}$$

and so $\tau = -r\hat{y}/K < 0$ and $\Delta = bc\hat{x}\hat{y} > 0$. Therefore $(\hat{x}, \hat{y})$ is a stable critical point and so it is either a stable node (perhaps degenerate) or a stable spiral point.

- (c) Write

$$\tau^2 - 4\Delta = \frac{r^2}{K^2} \hat{y}^2 - 4bc\hat{x}\hat{y} = \hat{y} \left[\frac{r^2}{K^2} \hat{y} - 4bc\hat{x} \right] = \hat{y} \left[\frac{r^2}{K^2} \hat{y} - 4b \frac{r}{K}(K - \hat{y}) \right]$$

using

$$c\hat{x} = \frac{r}{K}(K - \hat{y}) = \frac{r}{K} \hat{y} \left[\left(\frac{r}{K} + 4b \right) \hat{y} - 4bK \right].$$

Therefore $\tau^2 - 4\Delta < 0$ if and only if

$$\hat{y} < \frac{4bK}{\frac{r}{K} + 4b} = \frac{4bK^2}{r + 4bK}.$$

Note that

$$\frac{4bK^2}{r + 4bK} = \frac{4bK}{r + 4bK} \cdot K \approx K$$

where K is large, and $\hat{y} = a/b < K$. Therefore $\tau^2 - 4\Delta < 0$ when K is large and a stable spiral point will result.

21. The equation

$$x' = \alpha \frac{y}{1+y} x - x = x \left(\frac{\alpha y}{1+y} - 1 \right) = 0$$

implies that $x = 0$ or $y = 1/(\alpha - 1)$. When $\alpha > 0$, $\hat{y} = 1/(\alpha - 1) > 0$. If $x = 0$, then from the differential equation for y' , $y = \beta$. On the other hand, if $\hat{y} = 1/(\alpha - 1)$, $\hat{y}/(1 + \hat{y}) = 1/\alpha$ and so $\hat{x}/\alpha - 1/(\alpha - 1) + \beta = 0$. It follows that

$$\hat{x} = \alpha \left(\beta - \frac{1}{\alpha - 1} \right) = \frac{\alpha}{\alpha - 1} [(\alpha - 1)\beta - 1]$$

and if $\beta(\alpha - 1) > 1$, $\hat{x} > 0$. Therefore $(\hat{x}, \hat{y})$ is the unique critical point in the first quadrant. The Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} \alpha \frac{y}{y+1} - 1 & \frac{\alpha x}{(1+y)^2} \\ -\frac{y}{1+y} & \frac{-x}{(1+y)^2} - 1 \end{pmatrix}$$

and for $\mathbf{X} = (\hat{x}, \hat{y})$, the Jacobian can be written in the form

$$\mathbf{g}'((\hat{x}, \hat{y})) = \begin{pmatrix} 0 & \frac{(\alpha - 1)^2}{\alpha} \hat{x} \\ -\frac{1}{\alpha} & -\frac{(\alpha - 1)^2}{\alpha^2} - 1 \end{pmatrix}.$$

It follows that

$$\tau = - \left[\frac{(\alpha - 1)^2}{\alpha^2} \hat{x} + 1 \right] < 0, \quad \Delta = \frac{(\alpha - 1)^2}{\alpha^2} \hat{x}$$

and so $\tau = -(\Delta + 1)$. Therefore $\tau^2 - 4\Delta = (\Delta + 1)^2 - 4\Delta = (\Delta - 1)^2 > 0$. Therefore $(\hat{x}, \hat{y})$ is a stable node.

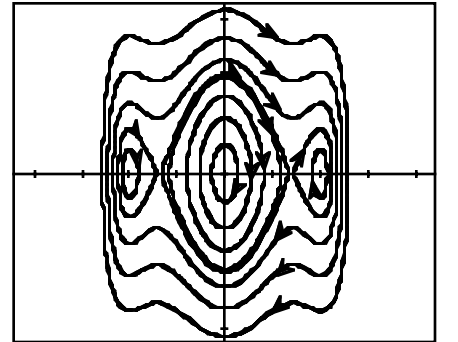
22. Letting $y = x'$ we obtain the plane autonomous system

$$\begin{aligned} x' &= y \\ y' &= -8x + 6x^3 - x^5. \end{aligned}$$

Solving $x^5 - 6x^3 + 8x = x(x^2 - 4)(x^2 - 2) = 0$ we see that critical points are $(0, 0)$, $(0, -2)$, $(0, 2)$, $(0, -\sqrt{2})$, and $(0, \sqrt{2})$. The Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ -8 + 18x^2 - 5x^4 & 0 \end{pmatrix}$$

and we see that $\det(\mathbf{g}'(\mathbf{X})) = 5x^4 - 18x^2 + 8$ and the trace of $\mathbf{g}'(\mathbf{X})$ is 0. Since $\det(\mathbf{g}'((\pm\sqrt{2}, 0))) = -8 < 0$, $(\pm\sqrt{2}, 0)$ are saddle points. For the other critical points the determinant is positive and linearization discloses no information. The graph of the phase plane suggests that $(0, 0)$ and $(\pm 2, 0)$ are centers.



EXERCISES 11.5

Periodic Solutions, Limit Cycles,
and Global Stability

1. $y' = x - y = 0$ implies that $y = x$ and so $2 + xy = 2 + x^2 > 0$. Therefore the system has no critical points, and so, by the corollary to Theorem 11.4, there are no periodic solutions.
2. $x' = 2x - xy = x(2 - y) = 0$ implies that $x = 0$ or $y = 2$. If $x = 0$, then from $-1 - x^2 + 2x - y^2 = 0$, $y^2 = -1$ and there are no real solutions. If $y = 2$, $-x^2 + 2x - 5 = 0$ which has no real solutions. Therefore the system has no critical points and so, by the corollary to Theorem 11.4, there are no periodic solutions.
3. For $P = -x + y^2$ and $Q = x - y$, $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = -2 < 0$. Therefore there are no periodic solutions by Theorem 11.5.
4. For $P = xy^2 - x^2y$ and $Q = x^2y - 1$, $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = y^2 - 2xy + x^2 = (y - x)^2$. Therefore $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$ does not change signs and so there are no periodic solutions by Theorem 11.5.
5. For $P = -\mu x - y$ and $Q = x + y^3$, $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = -\mu + 9y^2 > 0$ since $\mu < 0$. Therefore there are no periodic solutions by Theorem 11.5.
6. From $y' = xy - y = y(x - 1) = 0$ either $y = 0$ or $x = 1$. If $y = 0$, then from $2x + y^2 = 0$, $x = 0$. Likewise $x = 1$ implies that $2 + y^2 = 0$, which has no real solutions. Therefore $(0, 0)$ is the only critical point. But $\mathbf{g}'((0, 0))$ has determinant $\Delta = -2$. The single critical point is a saddle point and so, by the corollary to Theorem 11.4, there are no periodic solutions.
7. The corresponding plane autonomous system is $x' = y$, $y' = 2x - y^4$. Therefore $(0, 0)$ is the only critical point. But $\mathbf{g}'((0, 0))$ has determinant $\Delta = -2 < 0$. The single critical point is a saddle point and so, by the corollary to Theorem 11.4, there are no periodic solutions.
8. The corresponding plane autonomous system is

$$x' = y, \quad y' = -x + \left(\frac{1}{2} + 3y^2\right)y - x^2$$

and so $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 0 + \frac{1}{2} + 9y^2 > 0$. Therefore there are no periodic solutions by Theorem 11.5.

9. For $\delta(x, y) = e^{ax+by}$, $\frac{\partial}{\partial x}(\delta P) + \frac{\partial}{\partial y}(\delta Q)$ can be simplified to

$$e^{ax+by}[-bx^2 - 2ax + axy + (2b + 1)y].$$

Setting $a = 0$ and $b = -1/2$,

$$\frac{\partial}{\partial x}(\delta P) + \frac{\partial}{\partial y}(\delta Q) = \frac{1}{2}x^2e^{-\frac{1}{2}y}$$

which does not change signs. Therefore by Theorem 11.6 there are no periodic solutions.

11.5 Periodic Solutions, Limit Cycles, and Global Stability

10. For $\delta(x, y) = ax^2 + by^2$, $\frac{\partial}{\partial x}(\delta P) + \frac{\partial}{\partial y}(\delta Q)$ can be simplified to $-5ax^4 - 3bx^2y^2 + 10(a-b)x^2y$. Setting $a = b = 1$,

$$\frac{\partial}{\partial x}(\delta P) + \frac{\partial}{\partial y}(\delta Q) = -5x^4 - 3x^2y^2$$

which does not change signs. Therefore by Theorem 11.6 there are no periodic solutions.

11. For $P = x(1 - x^2 - 3y^2)$ and $Q = y(3 - x^2 - 3y^2)$,

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 4(1 - x^2 - 3y^2) \quad \text{and so} \quad \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} > 0$$

for $x^2 + 3y^2 < 1$. Therefore there are no periodic solutions in the elliptical region $x^2 + 3y^2 < 1$.

12. The corresponding plane autonomous system is $x' = y$, $y' = g(x, y)$ and so

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = \frac{\partial g}{\partial y} = \frac{\partial g}{\partial x'} \neq 0$$

in the region R . Therefore $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$ cannot change signs and so there are no periodic solutions by Theorem 11.5.

13. For $\delta(x, y) = \frac{1}{xy}$, $\delta P = -\frac{a}{y} + b$, $\delta Q = -c + \frac{r}{K} \frac{1}{x}(K - y)$ and so

$$\frac{\partial(\delta P)}{\partial x} + \frac{\partial(\delta Q)}{\partial y} = -\frac{r}{Kx} < 0$$

in the first quadrant. By Theorem 11.6, there are no periodic solutions in the first quadrant.

14. If $\mathbf{n} = (-2x, -2y)$,

$$\mathbf{V} \cdot \mathbf{n} = 2xy + 2x^2e^{x+y} - 2xy + 2y^2e^{x+y} = 2(x^2 + y^2)e^{x+y} \geq 0.$$

Therefore any circular region of the form $x^2 + y^2 \leq r^2$ is an invariant region by Theorem 11.7.

15. If $\mathbf{n} = (-2x, -2y)$,

$$\mathbf{V} \cdot \mathbf{n} = 2x^2 - 4xy + 2y^2 + 2y^4 = 2(x - y)^2 + 2y^4 \geq 0.$$

Therefore $x^2 + y^2 \leq r$ serves as an invariant region for any $r > 0$ by Theorem 11.7.

16. $\mathbf{n} = -\nabla t = (-6x^5, -6y)$. Since the corresponding plane autonomous system is $x' = y$, $y' = -y - y^3 - x^5$,

$$\mathbf{n} \cdot \mathbf{V} = -6x^5y + 6y^2 + 6y^4 + 6x^5y = 6y^2 + 6y^4.$$

Therefore $\mathbf{n} \cdot \mathbf{V} \geq 0$ and so by Theorem 11.7, the region $x^6 + 3y^2 \leq 1$ serves as an invariant region.

17. We showed in Example 8 that $\frac{1}{16} \leq x^2 + y^2 \leq 1$ is an invariant region for the plane autonomous system. If the only critical point is $(0, 0)$, this critical point lies outside the invariant region and so Theorem 11.8(ii) is applicable. There is at least one periodic solution in R .

18. The corresponding plane autonomous system is

$$x' = y, \quad y' = y(1 - 3x^2 - 2y^2) - x$$

and it is easy to see that $(0, 0)$ is the only critical point. If $\mathbf{n} = (-2x, -2y)$ then

$$\mathbf{V} \cdot \mathbf{n} = -2xy - 2y^2(1 - 3x^2 - 2y^2) + 2xy = -2y^2(1 - 2r^2 - x^2).$$

If $r = \frac{1}{2}\sqrt{2}$, $2r^2 = 1$ and so $\mathbf{V} \cdot \mathbf{n} = 2x^2y^2 \geq 0$. Therefore $\frac{1}{4} \leq x^2 + y^2 \leq \frac{1}{2}$ serves as an invariant region. By Theorem 11.8(ii) there is at least one periodic solution.

11.5 Periodic Solutions, Limit Cycles, and Global Stability

19. If $r < 1$ and $\mathbf{n} = (-2x, -2y)$ then

$$\mathbf{V} \cdot \mathbf{n} = -2xy + 2xy + 2y^2(1 - x^2) = 2y^2(1 - x^2) \geq 0$$

since $x^2 < 1$. Therefore $x^2 + y^2 \leq r^2$ serves as an invariant region. Now $(0, 0)$ is the only critical point and, since $\tau = -1$ and $\Delta = 1$, $\tau^2 - 4\Delta < 0$. Therefore $(0, 0)$ is a stable spiral point and so, by Theorem 11.9(ii), $\lim_{t \rightarrow \infty} \mathbf{X}(t) = (0, 0)$.

20. Since $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = -1 - 3y^2 < 0$, there are no periodic solutions. If $\mathbf{n} = (-2x, -2y)$,

$$\mathbf{V} \cdot \mathbf{n} = -2xy + 2x^2 + 2xy + 2y^4 = 2(x^2 + y^4) \geq 0.$$

Therefore the circular region $x^2 + y^2 \leq r^2$ serves as an invariant region for any $r > 0$. If (x, y) is a critical point, $y - x = 0$ or $y = x$. From $-x - y^3 = 0$ we have $-y(1 + y^2) = 0$. Therefore $y = 0$ and so $(0, 0)$ is the only critical point. It is easy to check that $\tau = -1$, $\Delta = 1$, $\tau^2 - 4\Delta = -3$ and so $(0, 0)$ is a stable spiral point. By Theorem 11.9(ii), $(0, 0)$ is globally stable. For any initial condition, $\lim_{t \rightarrow \infty} \mathbf{X}(t) = (0, 0)$.

21. (a) $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 2xy - 1 - x^2 \leq 2x - 1 - x^2 = -(x - 1)^2 \leq 0$. Therefore there are no periodic solutions.
 (b) If (x, y) is a critical point, $x^2y = \frac{1}{2}$ and so from $x' = x^2y - x + 1$, $\frac{1}{2} - x + 1 = 0$. Therefore $x = 3/2$ and so $y = 2/9$. For this critical point, $\tau = -31/12 < 0$, $\Delta = 9/4 > 0$, and $\tau^2 - 4\Delta < 0$. Therefore $(3/2, 2/9)$ is a stable spiral point and so, from Theorem 11.9(ii), $\lim_{x \rightarrow \infty} \mathbf{X}(t) = (3/2, 2/9)$.

22. (a) From $x \left(\frac{2y}{y+2} - 1 \right) = 0$, either $x = 0$ or $y = 2$. For the case $x = 0$, from $y \left(1 - \frac{2x}{y+2} - \frac{y}{8} \right) = 0$, $y \left(1 - \frac{y}{8} \right) = 0$. Therefore $(0, 0)$ and $(0, 8)$ are critical points. If $y = 2$, $1 - \frac{2x}{4} - \frac{1}{4} = 0$ and so $x = 3/2$.

Therefore $(3/2, 2)$ is the additional critical point. We may classify these critical points as follows:

$\mathbf{X}$	τ	Δ	$\tau^2 - 4\Delta$	Conclusion
$(0, 0)$	$-$	-1	$-$	saddle point
$(0, 8)$	$-$	$-\frac{3}{5}$	$-$	saddle point
$(\frac{3}{2}, 2)$	$\frac{1}{8}$	$\frac{3}{8}$	$-\frac{95}{64}$	unstable spiral point

- (b) By Theorem 11.8(i), since there is a unique unstable critical point inside the invariant region, there is at least one periodic solution.

CHAPTER 11 REVIEW EXERCISES

1. True
2. True
3. A center or a saddle point

4. Complex with negative real parts
5. False; there are initial conditions for which $\lim_{t \rightarrow \infty} \mathbf{X}(t) = (0, 0)$.
6. True
7. False; this is a borderline case. See Figure 11.25 in the text.
8. False; see Figure 11.29 in the text.
9. True
10. False; we also need to have no critical points on the boundary of R .
11. Switching to polar coordinates,

$$\begin{aligned}\frac{dr}{dt} &= \frac{1}{r} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right) = \frac{1}{r} (-xy - x^2 r^3 + xy - y^2 r^3) = -r^4 \\ \frac{d\theta}{dt} &= \frac{1}{r^2} \left(-y \frac{dx}{dt} + x \frac{dy}{dt} \right) = \frac{1}{r^2} (y^2 + xyr^3 + x^2 - xyr^3) = 1.\end{aligned}$$

Using separation of variables it follows that $r = \frac{1}{\sqrt[3]{3t + c_1}}$ and $\theta = t + c_2$. Since $\mathbf{X}(0) = (1, 0)$, $r = 1$ and $\theta = 0$. It follows that $c_1 = 1$, $c_2 = 0$, and so

$$r = \frac{1}{\sqrt[3]{3t + 1}}, \quad \theta = t.$$

As $t \rightarrow \infty$, $r \rightarrow 0$ and the solution spirals toward the origin.

12. (a) If $\mathbf{X}(0) = \mathbf{X}_0$ lies on the line $y = -2x$, then $\mathbf{X}(t)$ approaches $(0, 0)$ along this line. For all other initial conditions, $\mathbf{X}(t)$ approaches $(0, 0)$ from the direction determined by the line $y = x$.
- (b) If $\mathbf{X}(0) = \mathbf{X}_0$ lies on the line $y = -x$, then $\mathbf{X}(t)$ approaches $(0, 0)$ along this line. For all other initial conditions, $\mathbf{X}(t)$ becomes unbounded and $y = 2x$ serves as an asymptote.
13. (a) $\tau = 0$, $\Delta = 11 > 0$ and so $(0, 0)$ is a center.
- (b) $\tau = -2$, $\Delta = 1$, $\tau^2 - 4\Delta = 0$ and so $(0, 0)$ is a degenerate stable node.
14. From $x' = x(1 + y - 3x) = 0$, either $x = 0$ or $1 + y - 3x = 0$. If $x = 0$, then, from $y(4 - 2x - y) = 0$ we obtain $y(4 - y) = 0$. It follows that $(0, 0)$ and $(0, 4)$ are critical points. If $1 + y - 3x = 0$, then $y(5 - 5x) = 0$. Therefore $(1/3, 0)$ and $(1, 2)$ are the remaining critical points. We will use the Jacobian matrix

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 1 + y - 6x & x \\ -2y & 4 - 2x - 2y \end{pmatrix}$$

to classify these four critical points. The results are as follows:

$\mathbf{X}$	τ	Δ	$\tau^2 - 4\Delta$	Conclusion
$(0, 0)$	5	4	9	unstable node
$(0, 4)$	—	—20	—	saddle point
$(\frac{1}{3}, 0)$	—	— $\frac{10}{3}$	—	saddle point
$(1, 2)$	—5	10	—15	stable spiral point

If $\delta(x, y) = \frac{1}{xy}$, $\delta P = \frac{1}{y} + 1 - 3\frac{x}{y}$ and $\delta Q = \frac{4}{x} - 2 - \frac{y}{x}$. It follows that

$$\frac{\partial}{\partial x}(\delta P) + \frac{\partial}{\partial y}(\delta Q) = -\frac{3}{y} - \frac{1}{x} < 0$$

in quadrant one. Therefore there are no periodic solutions in the first quadrant.

CHAPTER 11 REVIEW EXERCISES

15. The corresponding plane autonomous system is $x' = y$, $y' = \mu(1 - x^2) - x$ and so the Jacobian at the critical point $(0, 0)$ is

$$\mathbf{g}'((0, 0)) = \begin{pmatrix} 0 & 1 \\ -1 & \mu \end{pmatrix}.$$

Therefore $\tau = \mu$, $\Delta = 1$ and $\tau^2 - 4\Delta = \mu^2 - 4$. Now $\mu^2 - 4 < 0$ if and only if $-2 < \mu < 2$. We may therefore conclude that $(0, 0)$ is a stable node for $\mu < -2$, a stable spiral point for $-2 < \mu < 0$, an unstable spiral point for $0 < \mu < 2$, and an unstable node for $\mu > 2$.

16. Critical points occur at $x = \pm 1$. Since

$$g'(x) = -\frac{1}{2}e^{-x/2}(x^2 - 4x - 1),$$

$g'(1) > 0$ and $g'(-1) < 0$. Therefore $x = 1$ is unstable and $x = -1$ is asymptotically stable.

17. $\frac{dy}{dx} = \frac{y'}{x'} = \frac{-2x\sqrt{y^2+1}}{y}$. We may separate variables to show that $\sqrt{y^2+1} = -x^2 + c$. But $x(0) = x_0$ and $y(0) = x'(0) = 0$. It follows that $c = 1 + x_0^2$ so that

$$y^2 = (1 + x_0^2 - x^2)^2 - 1.$$

Note that $1 + x_0^2 - x^2 > 1$ for $-x_0 < x < x_0$ and $y = 0$ for $x = \pm x_0$. Each x with $-x_0 < x < x_0$ has two corresponding values of y and so the solution $\mathbf{X}(t)$ with $\mathbf{X}(0) = (x_0, 0)$ is periodic.

18. The corresponding plane autonomous system is

$$x' = y, \quad y' = -\frac{\beta}{m}y - \frac{k}{m}(s+x)^3 + g$$

and so the Jacobian is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ -\frac{3k}{m}(s+x)^2 & -\frac{\beta}{m} \end{pmatrix}.$$

For $\mathbf{X} = (0, 0)$, $\tau = -\frac{\beta}{m} < 0$, $\Delta = \frac{3k}{m}s^2 > 0$. Therefore

$$\tau^2 - 4\Delta = \frac{\beta^2}{m^2} - \frac{12k}{m}s^2 = \frac{1}{m^2}(\beta^2 - 12kms^2).$$

Therefore $(0, 0)$ is a stable node if $\beta^2 > 12kms^2$ and a stable spiral point provided $\beta^2 < 12kms^2$, where $ks^3 = mg$.

19. For $P = 4x + 2y - 2x^2$ and $Q = 4x - 3y + 4xy$.

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 4 - 4x - 3 + 4x = 1 > 0.$$

Therefore there are no periodic solutions by Theorem 11.5.

20. If (x, y) is a critical point,

$$P = \epsilon x + y - x(x^2 + y^2) = 0$$

$$Q = -x + \epsilon y - y(x^2 + y^2) = 0.$$

But $yP - xQ = y^2 + x^2$. Therefore $x^2 + y^2 = 0$ and so $x = y = 0$. It follows that $(0, 0)$ is the only critical point and

$$\mathbf{g}'((0, 0)) = \begin{pmatrix} \epsilon & 1 \\ -1 & \epsilon \end{pmatrix}$$

so that $\tau = 2\epsilon$, $\Delta = \epsilon^2 + 1 > 0$, and so $\tau^2 - 4\Delta = -4$. Note that if $\mathbf{n} = (-2x, -2y)$, $\mathbf{n} \cdot \mathbf{V} = 2r^2(r^2 - \epsilon)$. If $\epsilon > 0$, then $\tau > 0$ and so $(0, 0)$ is an unstable spiral point. If $r = 2\epsilon$, $\mathbf{n} \cdot \mathbf{V} \geq 0$ and so $x^2 + y^2 \leq 4\epsilon^2$ is an invariant

region. It follows from Theorem 11.8(i) that there is at least one periodic solution in this region. If $\epsilon < 0$, $\tau < 0$ and so $(0, 0)$ is a stable spiral point. Note that $\mathbf{n} \cdot \mathbf{V} = 2r^2(r^2 - \epsilon) \geq 0$ for any r and so $x^2 + y^2 \leq r^2$ is an invariant region. It follows from Theorem 11.9(ii) that $\lim_{t \rightarrow +\infty} \mathbf{X}(t) = (0, 0)$ for any choice of initial position $\mathbf{X}_0$.

21. (a) If $x = \theta$ and $y = x' = \theta'$, the corresponding plane autonomous system is

$$x' = y, \quad y' = \omega^2 \sin x \cos x - \frac{g}{l} \sin x - \frac{\beta}{ml} y.$$

Therefore $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = -\frac{\beta}{ml} < 0$ and so there are no periodic solutions.

- (b) If (x, y) is a critical point, $y = 0$ and so $\sin x(\omega^2 \cos x - g/l) = 0$. Either $\sin x = 0$ (in which case $x = 0$) or $\cos x = g/\omega^2 l$. But if $\omega^2 < g/l$, $g/\omega^2 l > 1$ and so the latter equation has no real solutions. Therefore $(0, 0)$ is the only critical point if $\omega^2 < g/l$. The Jacobian matrix is

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ \omega^2 \cos 2x - \frac{g}{l} \cos x & -\frac{\beta}{ml} \end{pmatrix}$$

and so $\tau = -\beta/ml < 0$ and $\Delta = g/l - \omega^2 > 0$ for $\mathbf{X} = (0, 0)$. It follows that $(0, 0)$ is asymptotically stable and so after a small displacement, the pendulum will return to $\theta = 0$, $\theta' = 0$.

- (c) If $\omega^2 > g/l$, $\cos x = g/\omega^2 l$ will have two solutions $x = \pm \hat{x}$ that satisfy $-\pi < x < \pi$. Therefore $(\pm \hat{x}, 0)$ are two additional critical points. If $\mathbf{X}_1 = (0, 0)$, $\Delta = g/l - \omega^2 < 0$ and so $(0, 0)$ is a saddle point. If $\mathbf{X}_1 = (\pm \hat{x}, 0)$, $\tau = -\beta/ml < 0$ and

$$\Delta = \frac{g}{l} \cos \hat{x} - \omega^2 \cos 2\hat{x} = \frac{g^2}{\omega^2 l^2} - \omega^2 \left(2 \frac{g^2}{\omega^4 l^2} - 1 \right) = \omega^2 - \frac{g^2}{\omega^2 l^2} > 0.$$

Therefore $(\hat{x}, 0)$ and $(-\hat{x}, 0)$ are each stable. When $\theta(0) = \theta_0$, $\theta'(0) = 0$ and θ_0 is small we expect the pendulum to reach one of these two stable equilibrium positions.

- (d) In (b), $(0, 0)$ is a stable spiral point provided

$$\tau^2 - 4\Delta = \frac{\beta^2}{m^2 l^2} - 4 \left(\frac{g}{l} - \omega^2 \right) < 0.$$

This condition is equivalent to $\beta < 2ml\sqrt{g/l - \omega^2}$. In (c), $(\pm \hat{x}, 0)$ are stable spiral points provided that

$$\tau^2 - 4\Delta = \frac{\beta^2}{m^2 l^2} - 4 \left(\omega^2 - \frac{g^2}{\omega^2 l^2} \right) < 0.$$

This condition is equivalent to $\beta < 2ml\sqrt{\omega^2 - g^2/(\omega^2 l^2)}$.

22. The corresponding plane autonomous system is

$$x' = y, \quad y' = 2ky - cy^3 - \omega^2 x$$

and so $(0, 0)$ is the only critical point. Since

$$\mathbf{g}'(\mathbf{X}) = \begin{pmatrix} 0 & 1 \\ -\omega^2 & 2k - 3cy^2 \end{pmatrix}.$$

$\tau = 2k$ and $\Delta = \omega^2 > 0$. Since $k > 0$, $(0, 0)$ is an unstable critical point. Assuming that a Type I invariant region exists that contains $(0, 0)$ in its interior, we may apply Theorem 11.8(i) to conclude that there is at least one periodic solution.